Parasuppositions*

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Abstract. We introduce a new class of linguistic inferences, "parasuppositions", triggered by diverse constructions, notably appositive participles (e.g. *Ann drinks, taking enormous risks*); post-speech gestures—e.g. *Ann drinks – DRINK,* where *DRINK* is a gestural representation of drinking; and parenthetical conjuncts *q* of the form *p (and q)*—e.g. *Ann drinks (and takes enormous risks)*. With special reference to French, we show that parasuppositions give rise to projection patterns reminiscent of presuppositions: *Ann might smoke, taking enormous risks* yields the inference that *if Ann smokes, she takes enormous risks*. These projection patterns suggest that parasuppositions must be locally trivial (= entailed by their local context). But parasuppositions differ from presuppositions in two key respects: they are informative, and they are typically deviant under *not* and *none*-type quantifiers. While some appositive relative clauses (ARCs) might well trigger parasuppositions, our target constructions differ from ARCs in being preferentially interpreted in situ, and thus they do not display the "wide scope behavior" of ARCs. To explain the target behavior, we propose that parasupposition triggers must be entailed by their speaker's local context, computed on the basis of their linguistic environment combined with the speaker's beliefs (rather than with the common ground). We explain their deviance under *not* and *none* by the fact that in such environments they make a trivial contribution: *Nobody smokes, taking enormous risks* triggers two inferences, namely *Nobody smokes*, and *Everyone who smokes takes enormous risks*. But the second component is trivially true because of the first component, and the sentence has a simpler equivalent alternative, namely *Nobody smokes*. Long-term, our analysis raises questions about the typology of presupposition-like phenomena in language, and of their derivation.

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1 Introduction

1.1 Main goal: appositive participles

In the last 50 years, semantics has unearthed a rich typology of linguistic inferences that includes assertions (also called at-issue contributions), presuppositions, implicatures and supplements, among several others. We propose to add a new member to this interesting family, "parasuppositions", triggered by diverse constructions, notably appositive present participles (e.g. *Ann drinks, taking enormous risks* - henceforth APPs), but also post-speech gestures (e.g. *Ann drinks – DRINK,* where *DRINK* is a gestural representation of drinking) and parenthetical conjuncts *q* of the form *p (and q)* (e.g. *Ann drinks (and takes enormous risks))*.[1]

Parasuppositions resemble presuppositions in giving rise to specific patterns of projection under various operators, and more precisely to inferences that ensure that these constructions are locally trivial (= entailed by their local context). This can be illustrated in English by the parasupposition trigger *thus+present participle*, as in (1); the patterns of inference are identical to what would be found with a presupposition trigger <u>*p*</u>*p*' (with presupposition *p*), conjunctively added to *Ann smokes*, e.g. *she knows she is taking enormous risks*, as in (2).

(1) a. Does Ann smoke, thus taking enormous risks?
=> if Ann smokes, she takes enormous risks
b. Maybe Ann smokes, thus taking enormous risks.
=> if Ann smokes, she takes enormous risks
c. If Ann smokes, thus taking enormous risks, I won't be happy.
=> if Ann smokes, she takes enormous risks
d. If Ann is standing near the hydrogen generator, she is smoking, thus taking enormous risks.
=> if Ann is standing near the hydrogen generator, if she is smoking, she is taking enormous risks
e. Maybe every student smokes, thus taking enormous risks.
=> for every student x, if x smokes, x takes enormous risks

(2) a. Does Ann smoke and know she is taking enormous risks?
=> if Ann smokes, she takes enormous risks
b. Maybe Ann smokes and knows she is taking enormous risks
=> if Ann smokes, she takes enormous risks
c. If Ann smokes and knows she is taking enormous risks, she'll end up stopping.
=> if Ann smokes, she takes enormous risks
d. If Ann is standing near the hydrogen generator, she is smoking and knows she is taking enormous risks.
=> if Ann is standing near the hydrogen generator, if she is smoking, she is taking enormous risks
e. Maybe every student smokes and knows s/he is taking enormous risks.
=> for every student x, if x smokes, x takes enormous risks

Parasuppositions differ from presuppositions in two principal respects, however. First, unlike presuppositions, they must typically be informative, as illustrated by the deviance of (3).

(3) #Ann will travel, thus traveling.

Second, unlike presuppositions, parasuppositions are typically deviant under *not* and *none*-type quantifiers, as illustrated in (4) (we will see that this is not due to a 'wide scope behavior' of these constructions).

[1] Two terminological remarks should be added. The term 'parasuppositions' is intended to suggest that these suppositions are computed in parallel to assertions, and also to suggest that they share some but not all properties of presuppositions. Regarding APPs (Appositive Present Participles), they are obviously a special case of appositive participles, which can appear in the past tense. The latter can be expected to trigger parasuppositions as well, but in view of the diversity of the behavior of appositive participles summarized in Section 2.1, we prefer to concentrate on present participles in this piece.

(4) a. #Ann doesn't smoke, thus taking enormous risks.
b. #No student smokes, thus taking enormous risks

Phenomenally, then, parasuppositions are initially characterized by the triad in (5).

(5) **The parasuppositional triad**
a. Projection: Parasuppositions project like presuppositions.
b. Informativity: Parasuppositions must typically be informative (unlike standard presuppositions).
c. Deviance: Parasupposition triggers are typically degraded under negative expressions (unlike standard presupposition triggers).

To derive these properties, the key idea is that appositive participles must not be trivial relatively to their standard local context (hence their informativity requirement in (5)b), but must be trivial relative to what we may call their "speaker's local context", computed on the basis of the speaker's beliefs combined with the relevant linguistic environment (this derives the presupposition-like behavior in (5)a). The standard local context of an expression is computed of the context set, or in other words on what is taken for granted by the speaker and addressee, which is of course less specific than the speaker's beliefs. Correspondingly, the standard local context of an expression is usually less specific than its speaker's local context, and an appositive participle can be informative relative to the former while being trivial relative to the latter.

Crucially, when an appositive participle appears under negative expressions while projecting relative to the speaker's beliefs, it ends up making a vacuous contribution. To illustrate, consider (4)b. Rules of projection applied to the appositive participle give rise to the contribution that *every student that smokes takes enormous risks*. But this statement is made vacuously true and thus idle by the fact that *no student smokes*. More formally, if the speaker asserts (i) *[No N] [V, G]*, where *G* is a parasuppositional expression modifying the Verb Phrase *V*, rules of projection ensure that (ii) *[Every N] [V => G]*. But (i) and (ii) together are equivalent to a simpler sentence, namely (iii) *[No N] V*, as summarized in (6). The key idea, then, is that a parasuppositional expression is deviant in negative environments in which it could be eliminated without changing the information provided about the speaker's beliefs, something that violates a pragmatic condition of manner.

(6) a. Assertion: the speaker believes: (i) [No student] [V and G]
b. Projection: the speaker believes: (ii) [Every student] [V => G]
c. Alternative: the speaker could have said: (iii) [No student] V
d. Equivalence: [(i) & (ii)] <=> (iii)
=>: Suppose that (iii) is false and thus that for some x, V(x); then by (ii), G(x), and (i) couldn't be true.
<=: Suppose (iii). (i) immediately follows. (ii) does too because every student vacuously satisfies the material predicative conditional V => G.

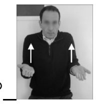
As we will see, this prohibition under *not* and *none* can be obviated when the parasuppositional expression plays a separate discourse role, thus supporting the idea that it is only to the extent that it is eliminable that the parasuppositional expression is deviant.

In its core, then, the theory to be developed can be summarized as in (7).

(7) **The core theory of parasuppositions**
a. Projection: a parasupposition triggered by π projects like a presupposition because it comes with a requirement that, relative to its speaker's local context s', it should satisfy: $s' \models \pi$
b. Informativity: a parasupposition triggered by π must typically be informative because it comes with a requirement that, relative to its standard local context c', it should satisfy: $c' \not\models \pi$
c. Deviance: a parasuppositional expression π modifying a VP is deviant under a negative expression Neg because, in these environments, *Neg … [VP, π]* ends up making the same informational contribution as *Neg… VP*, which is simpler.

1.2 Intuitive motivation: post-speech gestures

Why should there be presupposition-type conditions that constrain the context set, and others that constrain the speaker's beliefs? To provide intuitive motivation for our proposal, it might help to consider post-speech gestures, and their minimal difference relative to co-speech gestures (e.g. Schlenker 2018a,b). Post-speech gestures display striking similarities with appositive participles, and we will argue that they too trigger parasuppositions. First, they display the same projection patterns, as illustrated in (8). (Here and throughout, gestures are transcribed in capitals using a `non-standard font`, and sometimes suffixed with a picture that illustrates them.)

(8) a. Will Ann help her son – `UP_` ?
=> if Anne helps her son, lifting will be involved
b. Maybe Ann will help her son – `UP`.
=> if Anne helps her son, lifting will be involved
c. If Ann helps her son – `UP`, we will notice.
=> if Anne helps her son, lifting will be involved
d. If Ann's son wants to take the slides, will she help him – `UP`?
=> if Anne's son wants to use the slides, if she helps him, lifting will be involved
e. <> Maybe each of these dads will help his kid – `UP`.
=> for each of these dads x, if x helps x's kid, lifting will be involved

Second, they have to be informative, as shown by the deviance of (9)b, where the post-speech gesture fully repeats the content of the modifier *like this*.[2]

(9) #Ann will help her son like `UP_`this – `UP`.

Third, post-speech gestures are deviant under *not* and *none*, as seen in (10).

(10) a. #Ann won't help her son – `UP`.
b. One/#None of these 10 guys will help his son – `UP`.

In sum, post-speech gestures satisfy the phenomenological triad in (5).

On the basis of more partial data, recent research has argued that post-speech gestures trigger supplements and thus parallel appositive relative clauses (ARCs), not APPs. ARCs come with an informativity requirement (Potts 2005), and in addition give rise to inferences and distributional requirements that are reminiscent of those of post-speech gestures, as illustrated in (11)-(12).

(11) a. Will Ann help her son, which will involve lifting him?
=> if Anne helps her son, lifting will be involved
b. If Ann helps her son, which will involve lifting him, we will notice.
=> if Anne helps her son, lifting will be involved

(12) a. #Ann won't help her son, which will involve lifting him.
b. One/#None of these 10 guys will help his son, which will involve lifting him.

ARCs as analyzed as supplements were taken to only display a wide scope interpretive behavior (Potts 2005). As a result, the supplemental analysis of post-speech gestures never explained how the projection patterns in (5) follow in full generality. Only sample cases were considered, and had to rely on mechanisms of modal subordination (e.g. Roberts 1989). But closer inspection casts doubt on the supplemental analysis of post-speech gestures. In the counterfactual conditional in (13)a, the same projection patterns obtain as in other examples. In view of the "wide scope behavior" of ARCs, counterfactual mood (with *would*) must be used to obtain similar results in the ARC in (13)b. Counterfactual mood can similarly be used to derive the target inference in (13)c, which involves a

[2] The sentence in (i) is acceptable, but this is certainly because the gesture provides iconic information that enriches the lexical content of *lift*. In (9), by contrast, the post-speech gesture targets a modified VP that already has an iconic component (thanks *like `UP_`this*).

(i) Ann will help her son by lifting him – `UP`.

clausal parenthetical—a construction that uncontroversially displays a wide scope behavior (e.g. Schlenker 2023a).

(13) a. If Ann were to help her son – UP, we would notice.
=> if Anne helps her son, lifting would be involved
b. If Ann were to help her son, **which (i) #will / (ii) would involve lifting him**, we would notice.
c. If Ann were to help her son (it (i) #will / (ii) would involve lifting him), we would notice.
b(ii), c(ii) => if Ann were to help her son, lifting would be involved

The natural conclusion is that the post-speech gesture in (13)a is made acceptable by a covert version of counterfactual mood. But this assumption destroys the account of the deviance of post-speech gestures under *not* and *none*, as ARCs with *would* are acceptable in these environments, as shown in (14).

(14) a. Ann won't help her son, which would involve lifting him.
b. None of these 10 guys will help his son, which would involve lifting him.

To put the point differently, treating post-speech gestures as triggering supplements implies explaining away their projective behavior with powerful mechanisms of modal subordination; but these nullify the explanation of the deviance of post-speech gestures in negative environments. This observation motivates an alternative theory, namely the parasuppositional analysis of post-speech gestures.

Now post-speech gestures have the advantage of displaying a minimally different behavior from co-speech gestures, which have been argued to display a bona fide presuppositional behavior. As shown in the literature, if the examples in (8) are modified so as to include a co-speech rather than a post-speech gesture—with the lifting gesture UP co-occurring with *help*, the same projection patterns are found as in (8) (Schlenker 2018a). Crucially, however, the negative examples are much better than with post-speech gestures: there is a sharp acceptability contrast between (10) and (15).

Notation: When we transcribe co-speech gestures, they precede the expression they modify, enclosed within square brackets—e.g. UP_[help] transcribes a lifting gesture co-occurring with *help*.

(15) a. Ann won't UP_[help] her son.
=> if Ann helps her son, lifting will be involved
b. None of these 10 guys will UP_[help] his son.
=> for each of these 10 guys x, if x helps his son, lifting will be involved

Why should co-speech gestures and post-speech gestures project alike, while displaying a different behavior in negative environments? Regarding projection, both gestural types give rise to conditionalized presupposition-like inferences, e.g. *if Ann helps her son, lifting will be involved*. In presupposition theory, and more generally in theories of local contexts that underlie it (e.g. Heim 1983, Schlenker 2009), this requires that the gestural trigger should be computed after the VP it attaches to. This result is easy to derive for both gestural types because, irrespective of linear order, modifiers are computed (for purposes of local context computation) after the nominal and verbal elements they affect (Schlenker 2019, 2020). It is thus unsurprising that co-speech and post-speech gestures alike should be computed after the VPs they modify.

A deeper question is why these gestural types should display a presupposition-like behavior in the first place. It was speculated that co-speech gestures can easily be disregarded and should, for this reason, make a trivial contribution relative to their local context (Schlenker 2018a [fn. 50], 2018b, 2021b, 2023b). One could try to extend the same intuition to post-speech gestures, which can easily be severed from the constituents they modify. But it is currently unclear how one could motivate the idea that they are 'easy to disregard'.

Be that as it may, a key difference between co-speech and post-speech gestures is that only the latter have their own time slot. This makes them both more costly and more salient, and this justifies positing an informativity difference between the two gestural types: co-speech gestures needn't be informative, post-speech gestures must be. From there, it is natural to posit that co-speech gestures provide information about the context set, whereas post-speech gestures provide information about the speaker's beliefs. The standard co-speech requirement is that co-speech gestures must be entailed by their local context. The parasuppositional treatment of post-speech gestures proposes that they must follow from their speaker's local context (while being non-trivial in their standard local context), hence achieving the desired contrast between the two gestural types.

The minimal difference between co-speech and post-speech thus serves to motivate key components of the parasuppositional analysis.

1.3 Structure

The rest of this piece is organized as follows. We start by laying out the key target properties of APPs, with special reference to French, which has the advantage of offering minimal pairs between parasuppositional and at-issue participles (Section 2). We then develop an analysis of the informativity and projection requirements on parasuppositions (Section 3), before turning to their deviance under *not* and *none* (Section 4). We then derive predictions about further downward-monotonic environments (Section 5), and discuss the existence 'flip-flop effects' when negative expressions are stacked (Section 6). We then extend the parasuppositional analysis to two further constructions, parenthetical conjuncts (Section 7) and post-speech gestures (Section 8), before drawing some conclusions (Section 9).

2 Characteristic Properties of Parasuppositions: Appositive Present Participles

2.1 Appositive present participles: semantic variability and projection

APPs are often studied as part of a more general debate on "free adjuncts", predicative constructions without an overt subject that can be adjoined as appositives to various parts of a sentence. The traditional debate ("semantic variability") centers around two questions, primarily in English (Stump 1985). First, why does the construction *NP VP FA*, where *NP* is referential and *FA* is a free adjunct, sometimes entail *FA*, as in (16)a, and sometimes not, as in (16)b. In the first case the free adjunct is called strong, in the second it is called weak (and often gives rise to a conditional meaning).

(16) a. Being a master of disguise, Bill would fool everyone. (Strong free adjunct)
=> Bill is a master of disguise
b. Wearing that new outfit, Bill would fool everyone. (Weak free adjunct)
≠> Bill is wearing that new outfit
≈ if he wore that new outfit, Bill would fool everyone
(Stump 1985 p. 41)

The second question pertains to the variety of implicit relations that can connect the free adjunct to the clause it modifies, as illustrated in (17) for the case of strong adjuncts.

(17) a. Crossing the street, John was hit by a car.
Most likely interpretation: 'while he was crossing the street'
b. Crossing the street, John entered the bank.
Most likely interpretation: 'after he crossed the street'
c. Crossing the street, John entered a different county.
Most likely interpretation: 'by crossing the street'
(Stump 1985 p. 330)

In a corpus study, Kortmann 1991 (Tables 9.1, 9.2) finds multiple implicit relations, ranging from condition (for weak adjuncts), different temporal relations (anteriority, posteriority, simultaneity), cause, concession, contrast, etc.

Here we ask a different question: how do (some) postverbal APPs when they are embedded in complex sentences? As noted at the outset above, some APPs that immediately follow a VP give rise to projection patterns reminiscent of presuppositions. Our target pattern is illustrated in (18), with a comparison to presuppositional projection in (19). The conditional inferences obtained in the presuppositional case follow because in a sentence of the form *Maybe [p and* $\underset{\sim}{q}$*q']*, where $\underset{\sim}{q}$*q'* presupposes q (underlined), the entire sentence tends to presupposes *if p, q* (e.g. Heim 1983, Schlenker 2009).

(18) Projection with appositive participles
<> Maybe Ann entered the barn, startling a pony.
=> if Ann entered the barn, she startled a pony

(19) Projection with presuppositions
<> Maybe Ann entered the barn and regretted startling the newborn pony.
=> if Ann entered the barn, she startled the newborn pony

The appositive intonation (with a pause)is helpful to obtain such projection patterns. Without it, at-issue effects can be obtained, without projection, as in (20)

(20) Maybe Ann entered riding a pony.
≠> if Mary entered, she rode a pony

In view of their appositive character, these constructions would seem to fall under Potts's theory of supplements, but as we will see they display embedding possibilities that differ from those of appositive relative clauses (ARCs), one of the most salient examples of supplements (we return to this point in Section 2.3.2).

We hasten to add that, even with the restriction to postposed appositive present participles, there are many cases in which the appositive is arguably at-issue, as in (21), where the primed examples do not seem to us to give rise to projection.

(21) a. He fired, wounding one of the bandits. (Kortmann 1991 p. 151)
a'. Maybe he fired, wounding one of the bandits.
≠> if he fired, he wounded one of the bandits
b. So I wrote to Cram, suggesting a slightly alternative programme… (Kortmann 1991 p. 167)
b'. Maybe Ann wrote to Cram, suggesting a slightly alternative programme
≠>? if Ann wrote to Cram, she suggested a slightly alternative program

Our goal will not be to determine what triggers the projective use illustrated in (18), but to characterize the properties of the construction when this projective use is present.

2.2 *Key properties of appositive participles in French*

2.2.1 *Disambiguating appositive and restrictive uses*

The two uses of present participles illustrated in English in (18) (projective) and (20) (non-projective) are usually disambiguated by lexical (rather than just intonational) means in French—the at-issue use can be expressed with the participle preceded by *en* ('in') and without a pause; the appositive use can be expressed without *en*, and with a pause, a contrast illustrated in (22)-(23).

(22) a. Peut-être qu' Anne est entrée en chevauchant un poney.
maybe that Anne is entered in riding a pony
'Maybe Anne entered riding a poney.'
≠> if Mary entered, she rode a poney
b. Peut-être qu' Anne est entrée, chevauchant un poney.
maybe that Anne is entered riding a pony
'Maybe Anne entered, riding a poney.'
=> if Mary entered, she rode a poney

2.2.2 Projection patterns

As seen in (23), our target French APPs display the same patterns of projection as the English *thus*-participles mentioned in (2). The at-issue participles in (24) do not trigger such inferences: they display a narrow-scope at-issue behavior.[3]

(23) a. Est-ce qu' Anne voyage, prenant des risques énormes?
is-it that Anne travels, taking some risks enormous?
'Does Anne travel, taking enormous risks?'
b. Peut-être qu' Anne voyage, prenant des risques énormes.
maybe that Anne travels, taking some risks enormous.
'Maybe Anne travels, taking enormous risks.'
c. Si Anne voyage, prenant des risques énormes, je ne serai pas content.
if Anne travels, taking some risks enormous, I NE will-be not happy
'If Anne travels, taking enormous risks, I won't be happy.'
a, b, c => if Anne travels, she will take enormous risks
d. Si Anne devient correspondante à Damas, est-ce qu' elle voyagera, prenant
if Anne becomes correspondent in Damascus, is-it that she will-travel, taking
des risques énormes?
some risks enormous?
'If Anne becomes a correspondente in Damascus, will she travel, taking enormous risks?'
=> if Anne becomes a correspondent in Damascus, if she travels, she will take enormous risks
e. Peut-être que chaque étudiant voyage, prenant des risques énormes.
maybe that each student travels, taking some risks enormous.
'Maybe every student travels, taking enormous risks.'
=> for every student x, if x travels, x takes enormous risks.

(24) a. Est-ce qu' Anne voyage en prenant des risques énormes?
is-it that Anne travels en taking some risks enormous?
'Does Anne travel while taking enormous risks?'
b. Peut-être qu' Anne voyage en prenant des risques énormes.
maybe that Anne travels in taking some risks enormous.
'Maybe Anne travels while taking enormous risks.'
c. Si Anne voyage en prenant des risques énormes, je ne serai pas content.
if Anne travels in taking some risks enormous, I NE will-be not happy
'If Anne travels while taking enormous risks, I won't be happy.'
a, b, c ≠> if Anne travels, she takes enormous risks
d. Si Anne devient correspondante à Damas, est-ce qu' elle voyagera en prenant
if Anne becomes correspondent in Damascus, is-it that she will-travel in taking
des risques énormes?
some risks enormous?
'If Anne becomes a correspondente in Damascus, will she travel while taking enormous risks?'
≠> if Anne becomes a correspondent in Damascus, if she travels, she will take enormous risks
e. Peut-être que chaque étudiant voyage en prenant des risques énormes.
maybe that each student travels while taking some risks enormous.
'Maybe every student travels, taking enormous risks.'
≠> for every student x, if x travels, x takes enormous risks.

While one might initially think that the conditional inferences obtained ('if Ann travels, she will take enormous risks') solely involve the associate of the participle, this is not so. Thus in (24)e, the inference is not that if Anne travels (anywhere), she will take enormous risks, but more narrowly: if she becomes a correspondent in Damascus, if she travels, she will take enormous risks. Similarly, in (2)e the inference was not that if Ann is smoking, she is taking enormous risks, but more narrowly: if she is smoking near the hydrogen generator, she is taking enormous risks. This is all expected in view of standard rules of presupposition projection, as is illustrated in (2)e. The reason is that presuppositional (and now parasuppositional) requirements in the consequent of a conditional need not be satisfied by

[3] Most our glossing conventions are standard. Let us note the following conventions: *NE* serves to gloss *ne*, an expletive negative particle; *subj* = subjunctive marking; *fem* = feminine.

the global context, but rather by the local context of the consequent, which incorporates information about the antecedent (e.g. Heim 1983, Schlenker 2009).[4]

2.2.3 Narrow scope interpretation

ARCs have repeatedly been argued to display—obligatorily or preferentially, depending on the analysis—a wide scope semantic behavior, analyzed in semantic or syntactic terms depending on the theory (McCawley 1981, Potts 2005, Schlenker 2023a). APPs differ from ARCs in displaying (perhaps obligatorily) a narrow scope behavior. The contrast can be seen in (25). The ARC in (25)a can be interpreted outside the attitude verb—one needn't infer that Ann fears her son will increase his survival. But this inference is present with the APP in (25)b, which suggests that it interpreted in the scope of the attitude verb.[5]

(25) *Context:* Anne's son has a life-threatening disease.
a. Anne a peur que son fils prenne du Zopaclon, ce qui augmentera ses chances de survie.
Anne has fear that her son take of Zopaclon, it that will-incrase his chances of survival
'Anne fears her son is taking Zopaclon, which will increase his survival odds.'
≠> Anne fears her son might increase his survival odds
b. Anne a peur que son fils prenne du Zopaclon, augmentant ses chances de survie.
Anne has fear that her son take of Zopaclon, increasing his chances of survival
'Anne fears her son is taking Zopaclon, thus increasing his survival odds.'
=> Anne fears her son might increase his survival chances (hence pragmatic oddity)

The same contrasts are found in (26): with the ARC in (26)a, one needn't infer that Anne knows her son is putting his life at risk, but this inference is much stronger with the APP in (26)b.

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[4] This dependency on the antecedent of the conditional is sometimes (but not always) hard to detect owing to the Proviso Problem, which gives rise to strengthening of conditional presuppositions. See for instance Geurts 1996, 1999, Lassiter 2012, Mandelkern 2016.

[5] An additional argument could be investigated in the future. Schlenker 2023a uses the counterfactually interpreted imperfect to force narrow scope interpretation of ARCs under *if*. This test is used in (i) for APPs. For two of our consultants *but not all consultants*, the contrasts suggest that a clear narrow scope interpretation co-exists with a parasuppositional inference in (ia).

(i) *Context:* Trump has decided to send the national guard to California.
Si Trump était actuellement en Californie
if Trump was currently in California
'If Trump were currently in California

a. <ok> , se faisant insulter par toutes les personnes qu' il rencontrait,
, SE doing insult by all the pesons that he met,
, getting insulted by everyone he met,
b. <ok> et se faisait insulter par toutes les personnes qu' il rencontrait,
and SE did insult by all the pesons that he met,
and got insulted by everyone he met,
c. (#il se ferait insulter par toutes les personnes qu' il rencontrait),
(he SE would-do insult by all the pesons that he met),

il changerait d' avis.
he would-change of mind.
he would change his mind.'

(ic) establishes that the modally interpreted imperfect is degraded in matrix constructions, included if it is embedded under conditional mood. The APP in (ia) behaves like the conjunct in (ib) in taking narrow scope under *if* but, unlike it, triggers a parasuppositional inference. A detailed study of further consultants and of idiolectal variation would be needed to turn this into a genuine argument for narrow scope interpretation.

(26) *Context:* Anne's son is sick.
a. Anne sait que son fils prend du Zopaclon, ce qui met sa vie en danger.
Anne knows that her son takes of Zopaclon, it that puts his life at risk
'Anne knows that her son is taking Zopaclon, which puts his life at risk.'
≠> Anne knows her son is putting his life at risk
b. Anne sait que son fils prend du Zopaclon, mettant sa vie en danger.
Anne sait que son fils prend du Zopaclon, mettant sa vie en danger.
Anne knows that her son takes of Zopaclon, putting his life at risk
'Anne knows that her son is taking Zopaclon, putting his life at risk.'
=> Anne knows her son is putting his life at risk

2.2.4 Informativity

Potts's (2005) seminal work on supplements established that some non-at-issue elements can come with informativity requirements, as illustrated in (27).

(27) Lance Armstrong survived cancer.
a. #When reporters interview Lancer, a cancer survivor, he often talks about the disease. (Potts 2005).
b. #When reporters interview Lance, who survived cancer, he often talks about the disease.

Comparable effects are found with APPs, as shown in (28): clearly entailed APPs are degraded, as shown in (28)b,c; non-entailed APPs are acceptable, as shown in (28)a,d. By contrast, entailed contents are more acceptable if appearing in a conjunct combined with *donc* ('therefore'); this is unsurprising as *therefore* has been argued to require that its associate be presupposed to be entailed (Pavese 2017).

(28) Anne a voyagé en Syrie,
Anne has traveled to Syria,
a. prenant des risques énormes.
taking of risks enormous
taking enormous risks.'
b. (i) #voyageant. (ii) (?) et a donc voyagé.
traveling *and has therefore traveled*
(i) traveling.' (ii) and therefore traveled.'
c. (i)#se déplaçant. (ii) et s' est donc déplacée
SE moving *and SE is therefore moved*
(i) moving around.' (ii) and therefore moved around.'
d. se déplaçant beaucoup.
moving around a lot.'

Similar patterns are found in (29)-(30).

(29) a. #La semaine dernière, Anne a invité Marie et Jean, invitant Jean.
the week last Anne has invited Marie and Jean, inviting Jean
b. La semaine dernière, Anne a invité Marie et Jean,
the week last Anne has invited Marie and Jean,
et donc invité Jean.
and therefore invited Jean
'Last week, Anne invited Marie and Jean, and therefore she invited Jean.'

(30) *Context:* Ann is known to have a father and a mother.
a. #La semaine dernière, Anne a invité ses parents, invitant son père.
the week last Anne has invited her parents, inviting her father
b. La semaine dernière, Anne a invité ses parents, et donc invité son père.
the week last Anne has invited her parents, and therefore invited her father
'Last week, Anne invited her parents and therefeore invited her father.'

2.2.5 Deviance: downward-monotonic environments

Under *not* and *No P*, appositive participles are often deviant, as discussed for English in (4) (we will see below that there are discourse conditions that obviate this deviance). This is illustrated for French appositive participles in (31)-(32). (31)a has an irrelevant reading on which the participle modifies the entire negated VP, thus giving rise to the inference that Anne's *not* traveling makes her take enormous risks; the relevant reading is of course that on which Anne's traveling makes her take enormous risks.

(31) a. <#> Anne ne voyage pas, prenant des risques énormes.
Anne NE travels not, taking some risks enormous
b. #Aucun étudiant ne voyage, prenant des risques énormes.
no student NE travels, taking some risks enormous
a'. Anne voyage, prenant des risques énormes.
Anne travels, taking some risks enormous
'Anne travels, taking enormous risks.'
b'. Certains étudiants voyagent, prenant des risques énormes.
certain students travel, taking some risks enormous
'Some students travel, taking enormous risks.'

(32) Talking about a lottery:
a. #Anne n' a pas trouvé les bons numéros, gagnant 1000€.
Anne NE has not found the good numbers, wining 1000€
b. #Aucun joueur n' a trouvé les bons numéros, gagnant 1000€.
no player NE has found the good numbers, winning 1000€
a'. Anne a trouvé les bons numéros, gagnant 1000€.
Anne has found the good numbers, winning 1000€
'Anne found the winning numbers, thus winning €1000.'
b'. Plusieurs joueurs ont trouvé les bons numéros, gagnant 1000€.
several players have found the good numbers, winning 1000€
'Several players found the winning numbers, thus winning €1000.'

None-type operators come in different varieties. For instance, *it is impossible that p* means in essence that there is no accessible possible world in which *p* holds, and as expected this environment behaves like *none*, whereas *it is possible that p* behaves like *some*, as seen in (33).

(33) Talking about a lottery:
a. ??Il est impossible qu' Anne ait trouvé les bons numéros, gagnant 1000€.
it is impossible that Anne have found the good numbers, winning 1000€.
b. Il est possible qu' Anne ait trouvé les bons numéros, gagnant 1000€.
it is possible that Anne have found the good numbers, winning 1000€.
'It is possible that Anne found the winning numbers, thus winning €1000.'

With other downward-monotonic operators, the facts are more complex. With *less than 5,* the appositive participle is degraded in (34)a but not in (35)a, whereas the homologous examples with *none* in (31)b and (32)b are both degraded.

(34) a. ?? Moins de 5 étudiants ont voyagé, prenant des risques énormes.
less than 5 students have traveled, taking some risks enormous
b. Plus de 5 étudiants ont voyagé, prenant des risques énormes.
more than 5 students have traveled, taking some risks enormous
'More than 5 students traveled, thus taking enormous risks.'
c. Moins de 5 étudiants ont voyagé, ?et/ok mais ils ont pris des risques énormes.
less than 5 students have traveled, ? and/ok but they have taken some risks enormous
'Less than 5 students traveled, ?and / but they took enormous risks.'

(35) a. Moins de 5 joueurs ont trouvé les bons numéros, gagnant 10,000€.
fewer than 5 players have found the winning numbers, winning 10,000€
'Fewer than 5 players found the winning numbers, thus winning €10,000.'
b. Plus de 5 joueurs ont trouvé les bons numéros, gagnant 10,000€.

more than 5 players have found the winning numbers, winning 10,000€
'More than 5 players found the winning numbers, winning €10,000.'
c. Moins de 5 joueurs ont trouvé les bons numéros, et/mais ils ont gagné 10,000€.
fewer than 5 players have found the winning numbers, and/but they have won 10,000€
'Less than 5 players found the winning numbeeers, and/but they earned €10,000.'

We propose that (34)a is a bit degraded because the main clause and participle make contributions that are best connected by a discourse relation of contrast, as can be seen by the fact that *but* is a bit more natural than *and* in (34)c. When the discourse relation does not have to involve contrast, as in (35), the APP becomes more acceptable: the fewer winners there are, the larger the sum they can be expected to win, and thus there is no necessary adversative relation between the fact that fewer than 5 players found the winning numbers, and the fact that each won 10,000€.

Finally, other downward-monotonic environments seem to cause no problem for APPs, as seen in (36).

(36) a. Si Anne voyageait, prenant des risques énormes, ses parents seraient inquiets.
if Anne traveled, taking some risks enormous, her parents would-be worried.
'If Anne traveled, taking enormous risks, her parents would be worried.'
=> if Anne traveled, she would take enormous risks
b. Quand Anne voyage, prenant des risques énormes, ses parents sont inquiets.
when Anne travels, taking some risks enormous, her parents are worried.
'When Anne travels, taking enormous risks, her parents are worried."
=> when Anne travels, she takes enormous risks

2.2.6 Deviance: obviation through discourse relevance

Even with *not* and *none*, parasuppositions can be made acceptable if their contribution serves to justify the main clause, and probably more generally if the parasupposition fulfills certain discourse functions. This gives rise to apparent tense asymmetries, with appositive participles under *not* and *none* more acceptable in future than in past sentences, as illustrated in (37). For instance, (37)a is more acceptable than (37)b, but suggests that the *reason* the speaker thinks that none of his students will visit their parents is that this would unnecessarily put them at risk. We conjecture that past tense examples are less acceptable because this reading is less salient, as one likely has factual rather than inferential grounds for asserting that the student didn't visit their parents. A related contrast appears to hold between (37)a' and (37)b'.

(37) Uttered during the Covid pandemic:
a. (?) Mon étudiant ne va pas rendre visite à ses parents,
my student NE will not pay visit to his parents

leur faisant prendre des risques inutiles.
to-them making take some risks unnecessary
'My student won't visit his parents, thus unnecessarily putting them at risk.'
=> the reason the speaker thinks the student won't visit their parents is that this would put the latter at risk

b. ?? Mon étudiant n' a pas rendu visite à ses parents,
my student NE has not payed visit to his parents

leur faisant prendre des risques inutiles.
to-them making take some risks unnecessary

a'. (?) Aucun de mes étudiants ne va rendre visite à ses parents,
none of my students NE will pay visit to his parents

leur faisant prendre des risques inutiles.
to-them making take some risks unnecessary
=> the reason the speaker thinks his student won't visit their parents is that this would put the latter at risk

b'. ?(?)	Aucun	de mes	étudiants n'	a	rendre	visite	à	ses parents,
	none	*of my*	*students NE*	*will*	*payed*	*visit*	*to*	*his parents*

leur	faisant	prendre	des	risques	inutiles.
to-them	*making*	*take*	*some*	*risks*	*unnecessary*

We believe there are further factors that help make (37)a,a' more acceptable, notably intonation. We do not purport to give a complete analysis, but to suggest that the prohibition against APPs under *no* and *none* can be obviated, and that this comes with special discourse function for the appositive.

The fact that discourse relevance can obviate the prohibition against truth-conditionally redundant material is nothing new. It is discussed in relation with vacuous modifiers in Leffel 2014 (see also Schlenker 2005 for redundant modifiers in definite descriptions, and Esipova 2019 for a more general discussion of redundant modifiers). Specifically, Leffel proposes that *my sick mother* can be used when *sick* does not restrict the denotation of the NP (because the speaker has only one mother) if the mother's being sick enters in a discourse relation. Thus in (38)a, the mother's sick is the reason the speaker takes care of her, while (38)b is correspondingly deviant.

(38) a. I take care of my sick mother.
=> I take care of my mother because she is sick
b. #I take care of my sick mother, though her sickness has nothing to do with why.
(Leffel 2014, (3.61)b)

2.3 Comparing Parasuppositions with Presuppositions and ARC Supplements

For clarity, we summarize the similarities and differences between parasuppositions and presuppositions on the one hand, and supplements triggered by ARCs on the other. Importantly, we emphatically do not purport to distinguish parasuppositions from all appositive constructions, just from ARCs, for the simple reason that APPs *are* appositive constructions, and we suspect that several other appositive constructions trigger parasuppositions as well.

2.3.1 Parasuppositions versus presuppositions

While parasuppositions project like presuppositions, they differ from them in two respects: they are subject to an informativity requirement, and they are usually acceptable under *not* and *none*-type quantifiers.

We established the informativity requirement on APPs in Section 2.2.4. While presuppositions can often be informative (e.g. von Fintel 2008), they usually do not *have* to be, contrary to APPs.[6] Regarding monotonicity restrictions, standard presupposition triggers can be embedded under *not* and *none*, as illustrated in (39), although for *too* under negation this may require an additional level of syntactic embedding (Bade and Tiemann 2016; see also Beck 2006)[7].

(39) a. I will invite Ann, but my brother doesn't know it.
b. Yesterday I invited Ann, but tomorrow I won't invite her again.
c. I will invite Ann, but it's not the case that my brother will invite her too.

2.3.2 Parasuppositions versus ARC supplements

Parasuppositions resemble ARC-triggered supplements in being informative, as already illustrated in (27). But as we saw in Section 2.2.3, ARCs differ from APPs in being preferentially interpreted with maximally wide scope. The frequent wide scope behavior of ARC supplements makes it difficult to

[6] An interesting exception concerns redundant modifiers of definite descriptions, which must fulfill *some* function, which could be informativity. See for instance Schlenker 2005 and Leffel 2014 for discussion.
[7] Bade and Tiemann 2016 give the following contrast:
(i) Peter came to the party.
a. #John did not come to the party, too.
b. It is not the case that John came to the party, (too).

test monotonicity restrictions on ARCs. However, some tests have been developed to force narrow attachment of ARCs (Schlenker 2023). As illustrated in (40)a, one of them involves a subjunctive ARC in the scope of a modal ('possible'). The interpretation is very close to that of a conjunction embedded under the modal, as in (40)b: the truth conditions are compatible with a narrow scope behavior. Furthermore, licensing conditions on the subjunctive require this embedding, as shown by the deviance of (40)c.

(40) *Context:* There was a violent conflict at the Department.
a. (?)Il est possible que le gardien ait prévenu la directrice, qui ait
it is possible that the guard have-subj warned the director-fem, who has-subj

prévenu la doyenne.
warned the dean-fem.
'It is possible that the guard warned the director, who in turn warned the dean.'
≈ It is possible that the guard warned the director and that the latter warned the dean.
b. Il est possible que le gardien ait prévenu la directrice et qu'elle ait prévenu la doyenne.
it is

Il est possible que le gardien ait prévenu la directrice et qu'elle
it is possible that the guard have-subj warned the director-fem and that

ait prévenu la doyenne.
has-subj warned the dean-fem.
'It is possible that the guard called the director and that the latter warned the dean.'
c. *Il est possible que le gardien ait prévenu la directrice. Celle-ci ait *it is*
possible that the guard has-subj warned the director. That-one-fem has-subj
prévenu la doyenne.
warned the dean-fem.

Importantly, when such ARCs are forced to appear in the scope of *none*-type operators, deviance ensues. This is illustrated in (41)c, where we use the modal *impossible,* which means in essence *in no accessible possible world* and is thus a *none*-type operator, as argued in (33). The control sentence in (41)d is presumably a bit degraded because it triggers a slightly odd implicature ('it is possible that the guard warned the director or the latter warned the dean'). But to our ear, (41)c is worse.

(41) *Context:* There was a violent conflict at the Department.
a. (?) Il est possible que le gardien ait prévenu la directrice, qui ait
it is possible that the guard has-subj warned the director-fem, who has-subj
prévenu la doyenne.
warned the dean-fem.
'It is possible that the guard warned the director, who in turn warned the dean.'
b. Il est possible que le gardien ait prévenu la directrice et que
it is possible that the guard has-subj warned the director-fem and that prévenu
celle-ci ait prévenu la doyenne.
that-one-fem has-subj warned the dean-fem.
'It is possible that the guard warned the director and that the latter warned the dean.'
c. # Il est impossible que le gardien ait prévenu la directrice, qui ait
it is impossible that the guard has-subj warned the director-fem, who has-subj
prévenu la doyenne.
warned the dean-fem.
d. ? Il est impossible que le gardien ait prévenu la directrice et que
it is impossible that the guard has-subj warned the director-fem and that
celle-ci ait prévenu la doyenne.
that-one-fem has-subj warned the dean-fem.
'It is impossible that the guard warned the director and that the latter warned the dean.'

In view of such similarities between (narrow scope) ARCs and expressions that trigger parasuppositions, it might well be that ARC-triggered supplements should in the end be seen as a variety of parasuppositions. But their wide scope behavior sets them apart from the expressions under study in this piece.

2.4 Taking stock

In sum, we have seen that appositive participles have the properties in (42).

(42) **Properties of parasuppositional expressions**
a. Projection: They trigger inferences that project like presuppositions.
b. Informativity: They must be informative
c. Deviance: (i) They are usually deviant under *not* and *none*-type quantifiers. (ii) They are deviant under *less than n*-type quantifers, but only to the extent that controls involving separate sentences require *but*.
d. Obviation by discourse relevance: Deviance can be obviated if the (projected) parasuppositional inference fulfills certain discourse functions.

In the following, we will try to explain these properties, with the exception of (42)c(ii), which is left for future research.

3 Explaining Informativity and Projection

An analysis of parasuppositions must solve an apparent paradox. On the one hand, parasuppositions project like presuppositions, which might suggest that they are specified as being trivial in their local context. On the other hand, they are typically informative. These two constraints are incompatible if local triviality is computed relative to the standard local context, obtained from the context set C together with the linguistic environments (for rules of computation of local contexts, see for instance Heim 1983 and Schlenker 2009). The solution is to take parasuppositions to be computed from something stronger than C, call it C^+, with the effect that triviality in a local context obtained from C^+ needn't entail triviality relative to a local context computed from C. For simplicity, we initially take C^+ to be the speaker's beliefs S; we will briefly discuss alternative possibilities later.[8]

3.1 Basic analysis

In local context theory, the local context *c'* of an expression *E* plays a dual role (e.g. Stalnaker 1978, Schlenker 2009). First, it serves to verify that *E* makes a non-trivial contribution, in the sense that neither *E* nor its negation follows from *c'*, as is stated in (43). Second, if *E* triggers a presupposition π, then π should follow from *c'*, as is stated in (44).

(43) **Stalnaker's non-triviality conditions**
If E is an expression whose local context is c', then it should be the cas that:
a. $c' \not\models E$
b. $c' \not\models$ not E

(44) **Stalnaker's presupposition satisfaction condition**
If E is an expression whose local context is c', and if E triggers a presupposition π, then it should be the case:
$c' \models \pi$

Crucially, we propose that two notions of local context enter in the licensing of parasuppositions. We posit that, before projection, parasupposition triggers have a conjunctive semantics. Thus *x travels, taking risks* means: *x travels and x takes risks*. We assume that a parasuppositional expression π is subject to standard non-triviality conditions: its local context *c'*, computed from the context set *C* together with the linguistic environment of π should neither entail π

[8] Schlenker 2023 took narrow scope ARCs to be subject to the condition that they should be (i) informative in their local context computed from C, (ii) trivial in their local computed from a strengthened context set C^+, where C^+ is obtained from C by adding to it assumptions that are 'easy to accommodate'. This earlier analysis is thus compatible with the present one.

nor its negation, as is stated in (45)c. But in addition, π should display the behavior of a presuppositional expression in being entailed by its local context—not in the sense of the standard local context *c*', but rather in the sense of the speaker's local context *s*' computed from the speaker's belief set *S*, as is stated in (45)d. Clearly, if a proposition follows from *C*, then it follows from *S* as well, but the converse is not true: the speaker may for instance believe a proposition that the addressee doesn't believe, in which case this proposition follows from *S* but not from *C*. In other words, *S* is a smaller set than *C*, and correspondingly a speaker's local context *s*' derived from *S* and a certain linguistic environment will be at least as strong as a standard local context *c*' derived from *C* and the same linguistic environment.[9]

(45) Let π be a parasuppositional expression. We write [VP * π] for a VP followed by π.[10]
a. Target: … DP [VP * π]…
is uttered relatively to
a context set C
a speaker's belief set S
b. The literal meaning of [VP * π] is the predicative conjunction of VP and π.
c. Informativity condition
π should not be trivial relative to its local context c' computed from C together with the linguistic environment that precedes π:
$c' \not\models \pi$
$c' \not\models$ not π
d. Projection condition
π should be follow from its speaker's local context s', computed from S together with the linguistic environment that precedes π:
$s' \models \pi$

3.2 Illustrations

Let us illustrate the main notions with some examples.

In (46), we consider a simple unembedded example, such as *Ann smokes, taking enormous risks* (or *#Ann invited her parents, inviting her father*, for a deviant version). Its general form is *A [VP * π]*, with π the parasuppositional expression, as in (46)a'''. Its literal meaning is that of a conjunction, as in (46)b. The non-triviality condition in (46)c(i) requires that it shouldn't be presupposed relative to the context set that *if Ann smokes, she takes enormous risks*. The violation of this condition explains the deviance of *Ann invited her parents, inviting her father*. The projection condition in (46)c is that the speaker believes the material conditional *if Ann smokes, she takes enormous risks*. But this is trivially satisfied if the speaker believes the content, before projection, of the utterance, namely *Ann smokes and takes enormous risks*.

(46) **Unembedded case**[11]
We call *c*' the local context of π (computed relative to the context set *C* and the linguistic environment of

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[9] Here we assume a definition of local contexts similar to Schlenker 2009, 2011, as in (i). Here *c*' is the strongest element that can be added in a formula while guaranteeing that certain equivalences hold throughout *C*. It is clear that if *C* is replaced with a smaller set C^+, this requirement will become easier to satisfy, and thus the local context c^{+}' computed from C^+ will be at least as strong as the local context *c*' computed from *C*.

(i) The local context of an expression *d* of propositional or predicative type which occurs in a syntactic environment *a _ b* in a context C is the strongest proposition or property x which guarantees that for any expression *d'* of the same type as *d*, for all strings *b'* for which *a d' b'* is a well-formed sentence,

$C \models^{c' \to x}$ a (c' and d') b' $\leftrightarrow$ a d' b'
(Schlenker 2011)

[10] We write [VP * π] rather than [VP, π] because our analysis is intended to apply to constructions besides APPs.
[11] We use the following standard notations (related to those of Schlenker 2009):
x is the type of individuals, s is the type of worlds, t is the type of truth values
Propositional expressions are of type <s, t>, predicates are of type <s, <e, t>>.
E^w is the value of expression E at world w

π), and s' the speaker's local context of π (computed relative to the speaker's belief set S and the linguistic environment of π).
a. Ann smokes, taking enormous risks.
a'. #Ann invited her parents, inviting her father.
a". A [VP * π]
b. Literal meaning
$w \models$ A [VP * π] iff $w \models$ A [VP and π]
c. Non-triviality conditions
The local context of π is: $c' = \lambda w\ \lambda x\ \ C(w) = 1$ and x = Ann and $\mathbf{VP}^w(x) = 1$. Conditions:
(i) $c' \not\models \pi$
(ii) $c' \not\models$ not π
hence[12]
$[\lambda w\ \ C(w) = 1$ and $\mathbf{VP}^w(\text{Ann}) = 1] \not\models [\lambda w\ \pi^w(A)]$
(This accounts for the deviance of a'.)
c. Projection condition
The projection condition is vacuous as soon as the speaker is in a position to assert the literal meaning of the construction:
$S \models$ A [VP and π]
hence in particular
$S \models \pi(A)$
It immediately follows that if s' is the local context of π, $s' \models \pi$

We can apply the same reasoning to a more complex case, involving embedding under a modal, as in (47)a. We obtain a non-triviality condition: it should not be presupposed that *if Ann smokes, she takes enormous risks*. But the projection condition is now more interesting, as it genuinely enriches the meaning obtained before projection: throughout the domain of the existential modal, the material implication *if Ann smokes, she takes enormous risks* should hold—a much stronger condition than is imposed by: *it might be that Ann smokes and takes enormous risks*.

(47) **Embedding under *might***
a. Ann might smoke, taking enormous risks.
a'. #Ann might invite her parents, inviting her father.
a". Might A [VP * π]
b. Literal meaning
$w \models$ Might A [VP * π] iff $w \models$ Might A [VP and π]
c. Non-triviality conditions
With the same value for c' as in (46)b,
(i) $c' \not\models \pi$
(ii) $c' \not\models$ not π
c. Projection condition
The value s' of the speaker's local context of π is:
$s' = \lambda w\ \ \lambda x\ \ S(w) = 1$ and x = Ann and $\mathbf{VP}^w(x) = 1$
hence
$\lambda w\ \ S(w) = 1$ and $\mathbf{VP}^w(\text{Ann}) = 1 \models \pi(A)$
hence: the speaker believes that if Ann VPs, Ann π's,
i.e. for a.: the speaker believes that if Ann smokes, she takes enormous risks.

Using local context theory (e.g. Schlenker 2009), the same analysis can be applied to quantified examples such as (23)e (*Maybe every student smokes, thus taking enormous risks*).

If w is of type s (the type of worlds) and E is an expression, $w \models E$ means that E is true at w.
If C is of type <s, t> and E is an expression, $C \models E$ means that for all worlds w such that $C(w) = 1$, $w \models E$.
If c is of type <s, <e, t>> and E is an expression, $c \models E$ means that for all worlds w and individuals x such that $c(w)(x) = 1$, $E^w(x) = 1$.
w ranges over worlds, x over individuals.
[12] $c' \models \pi$ can be rewritten as in (i)a below. Writing A for Ann, it is equivalent to (i)b and to (i)c.
(i) a. for all individuals x, for all worlds w, $[C(w) = 1$ and x = Ann and $\mathbf{VP}^w(x) = 1] \Rightarrow \pi^w(x) = 1$
b. for all worlds w, $[C(w) = 1$ and $\mathbf{VP}^w(A) = 1] \Rightarrow \pi^w(A) = 1$
c. $[\lambda w\ C(w) = 1$ and $\mathbf{VP}^w(A) = 1] \models [\lambda w\ \ \pi^w(A)]$

3.3 Refinements

What matters to reconcile projection and informativity is that the former should be computed relative to C^+ rather than C, the standard context set, with C^+ stronger than C (i.e. $C^+ \subset C$). Setting $C^+ = S$ is one way to achieve this result, but it is not the only one. A salient alternative is that C^+ is CS, what it is presupposed that the speaker believes.[13] This is of course different from what is simply presupposed, as is illustrated in (48). (48)a is appropriate even if the debate hasn't been adjudicated. (48)b is only appropriate if the addressee has conceded, otherwise the sentence is humorous or polemical.

(48) *Context:* The speaker and addressee had a public argument at a conference a year ago.
a. Does this journalist know that I believe that you were wrong?
b. Does this journalist know that you were wrong?

On the usual assumption that the speech act participants do not have false beliefs, it is clear that in general $S \subset CS$ ($\subset C$). Thus if parasuppositions are required to be satisfied in their local context computed relative to CS rather than S, we will have a harder requirement to fulfill.

We leave open whether S, CS or some other value of C^+ would be best to analyze the epistemic status of parasuppositions.

4 Explaining Deviance under Negative Expressions

4.1 A manner-based analysis: Non-eliminability

Why are parasuppositional expressions degraded under negative expressions? The basic idea is this. Suppose the speaker asserts a sentence … *[P * π]* … containing a predicate *P* and a parasupposition trigger *π*. On Gricean grounds of manner, it should not be common belief that the truth-conditional effect of the assertion is the same as if the same sentence had been uttered without *π*, unless *π* fulfills some other function, as stated in (49).[14]

(49) **Non-eliminability condition (initial version)**
If … [P * π] … is asserted by a speaker whose belief state is S,
a. it should not be (be common belief) that S |= … [P * π] … <=> S |= … P …
b. … unless π fulfills certain other discourse functions.

[13] In standards characterizations (used for instance in Schlenker 2012), C can be seen as the set of worlds reachable from the actual world through a combination of 1-steps and 2-steps, where a 1-step corresponds to the speaker's accessibility relation, and a 2-step to the addressee's accessibility relation. In this characterization, *CS* is the set of worlds reachable from the actual world through a path that ends in a 1-step, as detailed below. It is clear that in general $CS \subset C$.

(i) Let R_1 be the epistemic accessibility relation for the speaker and R_2 the epistemic accessibility relation for the addressee.
w' is reachable from w just in case some path π leads from w to w' through a series of steps of the form $<w_i, w_{i+1}>$ for which either $w_i R_1 w_{i+1}$ or $w_i R_2 w_{i+1}$. In the former case, we say that $<w_i, w_{i+1}>$ is a 1-step of π, and in the latter case we say that $<w_i, w_{i+1}>$ is a 2-step of π.

(ii) At w, it is common belief that *F* just in case *F* holds in every world reachable from w.

(iii) At w, it is common belief that the speaker believes that *F* just in case *F* holds in every world reachable from w through a path that ends in a 1-step (applying (ii) to: it is common belief that *SF*, where *SF* is: the speaker believes that *F*).

[14] We distinguish between "truth-conditional effect" and "other function" for simplicity, but in many or all cases, the "other function" is in fact a more complex truth-conditional effect involving some discourse relations. For instance, in (37)a, the final truth conditions are of the rough form: *My student won't visit his parents because this would unnecessarily put them at risk.*

This constraint follows from standard Gricean conditions of manner, according to which prolixity should be avoided unless it has an effect. It also followtruts from Katzir's (2007) theory of alternatives, according to which an alternative which is equivalent but strictly simpler than the sentence uttered is strictly better than it (I).[15]

Now when a sentence ... *[P * π]* ... with a parasuppositional expression π is uttered, two things immediately become common belief (for simplicity, we disregard the non-triviality condition on parasuppositions):
(a) the speaker believes ... *[P * π]* ...;
(b) the speaker's belief is such that $P \Rightarrow \pi$ follows from the speaker's local context *s'* of *P*.
If we write, as before, S |= F for *the speaker believes F*, we can state these two informational results as in (50).[16]

(50) If the speaker asserts a sentence ... *[P * π]* ... containing a predicate *P* and a parasupposition trigger π, then the addressee acquires the following information after assertion:
a. Assertive condition
S |= ... [P and π] ...
(= the speaker believes the meaning of the sentence before projection of the parasupposition)
b. Projective condition
If s' is the local context of P relative to S, s' |= P => π
(= for the speaker, the projective condition is satisfied)

To derive deviance under *not* and *none*, it is enough to show that, had the speaker *not* included the parasuppositional expression, the meaning would not have been affected, hence a violation of Gricean manner.

Let us first consider negation, as in *Ann doesn't [smoke, thus taking enormous risks]*. Let us analyze it for simplicity in propositional rather than predicative terms, as: *not [p, g]*, with *p = Ann smokes*, and *g = Ann takes risks*. This gives rise to the two informational contributions in (51):

(51) a. Assertive condition: S |= not [p and π]
b. Projective condition: If s' is the local context of p relative to S, s' |= p => π, and since s' is just S: S |= p => π

In sum, the informational contribution of the sentence, i.e. what it makes it common belief that the speaker believes, is the conjunction in (52)a. But this is the very same contribution made by *not p*, which is obtained from *not [p, g]* by removing the parasupposition trigger, as shown in (52)c.

(52) a. S |= not [p and g] and [p => g]
b. S |= not p
c. For any S, (a) iff (b)
(a) => (b): Suppose (a) and suppose, for contradiction, that for some w in S, w |≠ not p, i.e. w |= p. By (a), w |= g. But then it cannot be that S |= not [p and g]
(b) => (a): immediate

Let us now extend the reasoning to *none*-type examples, for instance *#None of my students smokes, thus taking enormous risks,* analyzed in schematic form in (53).

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[15] In a nutshell, Katzir 2007 takes a sentence *S* to be at least as good as a sentence *S'* just in case (i) *S* is at least as structurally simple as *S'*, and (ii') *S* entails *S'* (see Katzir's (23)). (i) is cashed out in terms of *S* being derivable from *S'* through a finite series of deletions, contractions, and replacements of constituents (see Katzir's (19)). It is clear that for a sentence *S' = Ann doesn't smoke, taking enormous risks,* the structurally simpler sentence *S = Ann doesn't smoke* is at least as good as *S'* when *S* and *S'* are equivalent. Furthermore, *S'* is not as good as *S* because it cannot be derived from *S* through a finite series of deletions, contractions and (reasonable) replacements of constituents. Therefore *S* is strictly better than *S'*.

[16] Note that we could state (50)b in a more roundabout way, by first requiring that the speaker's local context *s"* of π entail it, and then noting that, in view of the conjunctive semantics of $P * \pi$, this boils down to the requirement in (50)b.

(53) Suppose a speaker with beliefs S asserts a sentence S_1 = [No N] [V * π]…, and let s' be the speaker's local of π. The simpler alternative without the parasupposition trigger is the sentence S_2 = [No N] V
a. For S_1, the assertive condition before projection yields:
S $\models$ [No N] [V and π]
b. For S_1, the projective condition on parasuppositions yields:
s' $\models \pi$
c. For S_1, standard rules of local context computation guarantee that b. is equivalent to:
S $\models$ [Every N] [V => π]
d. In sum, S_1 provides the following information about S:
S $\models$ [No N] [V and π] and [Every N] [V => π]
e. The simpler alternative S_2 yields the information:
S $\models$ [No N] V
f. The informational contents in d. and e. are identical:
d => e: Suppose that d, and suppose, for contradiction, that for some world w of S, w $\not\models$ [No N] V, and thus w $\models$ [Some N] V. By the second conjunct in d, w $\models$ [Some N] [V and π], but this contradicts the first conjunct of e.
e => d: immediate
g. Since the simpler alternative S_2 yields the same informational content as the more complex parasuppositional sentence S_1, the latter is deviant, unless its parasuppositional component is justified by some other pragmatic conditions.

One could ask whether there are further operators that make parasupposition triggers eliminable and thus deviant. On the basis of the properties above, two results can be derived (see Appendix I for details). If a propositional operator Op yields the same type of parasupposition eliminability as negation, then it must entail negation (with the additional assumption that Op doesn't *always* yield falsity, it must be identical to negation). If a generalized quantifier *D P* (with D a determiner and P a predicate) yields the same type of parasupposition eliminability as *No P*, then it must entail *No P*.[17] As we noted in (36), APPs are acceptable in other downward-monotonic environments such as *if*- and *when*-clauses. And as discussed in Section 2.2.5, under *less than n*-type quantifiers, deviance is not systematic and, when it obtains, it might be due to other reasons (although this will require further investigation).

4.2 Obviating deviance under negative expressions

It is in very spirit of a manner condition that an expression that does not contribute to the literal meaning of the sentence might still be included if it fulfills some other function, as in (37)a above. In 'My student won't visit his parents, thus unnecessarily putting them at risk.', the conditional *if my student visits his parents, he will unnecessarily put them at risk* served to explain why my student won't visit his parents. This can be compared to (54)a and especially (54)b below, which has the advantage of having a simple propositional semantics. It is clear that the literal meaning of (54)b is equivalent to that of (54)c. Still, the more complex construction is not blocked by the simpler one because the former conveys that the speaker's grounds for their conclusion is that they believe the conditional.

(54) a. If my student visits his parents, he will put them at risk. So he won't visit his parents..
b. Either my student won't visit his parents or he will put them at risk. So he won't visit his parents.
c. My student won't visit his parents.

It is a separate enterprise to explain how discourse relations can achieve this kind of enriched meaning in our examples with APP (see Leffel 2014 for relevant analyses pertaining to vacuous modifiers). But it is clear that such an enriched meaning suffices to explain why Non-eliminability can be obviated.

[17] Note that it could still make additional demands. For instance, in a bivalent framework (which blurs the distinction between presupposition failure and falsity), *None of the five students Q* is true just in case (i) there are five students, and (ii) no student Qs. This entails *No student Q*, and thus yields parasupposition eliminability.

4.3 Relation to recent analyses of some PPIs

Our theory bears a resemblance to a recent analysis of some PPIs (Positive Polarity Items). And in fact, parasupposition triggers are, descriptively, weak PPIs, a point we return to in Section 6.4 (they are 'weak' because they are degraded under negation and *none*-type operators [van der Wouden 1997, Penka 2020]).[18] Now Solt and Wilson 2021 and Nicolae and Solt 2025 argue that some PPI's, such as *fairly* in (55), are degraded under negation due a constraint of manner.

(55) a. Aliona is fairly tall.
a'. Aliona is tall.
a". Aliona is very tall/
b. *Aliona isn't fairly tall.
b'. Aliona isn't tall.
(Nicolae and Solt 2025)

In a nutshell, their key idea is that *fairly* has such a weak semantics that *fairly tall* is equivalent to *tall*. By Katzir's theory of alternatives (Katzir 2007), (55)b is ruled out because (55)b' is equivalent but simpler. But if so, why is (55)a acceptable? While (55)a' is equivalent and simpler, (55)a but not (55)a' triggers the implicature that *Aliona isn't very tall*, through competition with (55)a". Applying once again Katzir's theory of alternatives, the reason is that the sentence with *is fairly tall* automatically evokes the sentence with *is very tall* (replacing *fairly* with *very*), but *is tall* doesn't evoke *is very tall*, which is strictly more complex. In the end, the truth-conditional contribution of (55)a with the implicature is different from that of the simpler sentence in (55)a'. This explains why no manner condition is violated.

Our analysis of the distribution of parasupposition triggers differs because we predict a contrast between positive and negative environment without appealing to implicatures. On the other hand, we will see in Section 5.2 that implicatures could play an explanatory role when more complex examples are considered.

4.4 Relation between Non-eliminability and Stalnaker's non-triviality conditions

The Non-eliminability condition in (49)a looks very different from Stalnaker's non-triviality condition in (43)a ($c' \not\models E$, i.e. the local context of an expression E shouldn't entail it). The former prohibits truth-conditional eliminability in a sentence, whereas the latter prohibits entailment by a local context. In addition, the Non-eliminability condition is evaluated relative to the speaker's beliefs, whereas Stalnaker's condition is evaluated relative to the context set. In Appendix II, we show that (i) the Non-eliminability condition in (49) is not reducible to Stalnaker's non-triviality conditions, no matter which context they are evaluated with respect to, but that (ii) one half of Stalnaker's conditions, namely (43)a, follows from the Non-eliminability condition.

4.5 Taking stock

In sum, positing that parasupposition triggers must be entailed by the speaker's local context has several benefits. It explains why they project like presuppositions, and yet make non-trivial contributions (relative to their standard local context). Furthermore, on the basis of standard conditions of manner (as incorporated for instance in Katzir 2007), the analysis can explain why parasupposition triggers are degraded under *not* and *none*, and why these cases can be ameliorated when the parasupposition fulfills some other discourse function.

An important caveat is in order. We suggested in Section 3 that parasuppositional projection and non-triviality can be reconciled if projection (i.e. entailment of the parasupposition by its local context) is computed relative to C^+ rather than C, the standard context set, with C^+ stronger than C. But in Section 4.1, we derived deviance under negative expressions by assessing projection relative to the speaker's beliefs *S,* with the key steps in (56) for the case of an unembedded negation.

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[18] If one disagrees with our analysis of the deviance of some (though not all) parasupposition triggers under quantifiers such as *less than n*, one might want to revise the claim that they are *weak* NPIs. (Note that if one takes them to be strong PPIs, one would still have to explain why parasupposition triggers are acceptable in *if*-clauses, which are downward-entailing environments.)

(56) a. Assertive condition: S |= not [p and g]
b. Projective condition: S |= p => g
c. (a) and (b) is equivalent to: S |= not p

This reasoning goes through with $C^+ = S$, i.e. if projection is evaluated relative to the speaker's beliefs. But the reasoning won't go through if C^+ is a strict superset of S, for in this case the projective condition ($p => g$) will not be made vacuous by the assertive condition of the simpler alternative (*not p*), as is illustrated in (57).

(57) A case in which the projective condition is not redundant
a. Assertive condition: w |= not [p and g]
b. Projective condition: {w, w'} |= p => g
c. (a) and (b) is NOT equivalent to: w |= not p
as the latter does not imply that: w' |= p => g

If one were to explore such refinements regarding the epistemic status of the projective condition (with $C^+ \neq S$), one might have to adapt the account of deviance as well.[19]

5 Further Predictions: Embedding Conjunctions

We turn to fine-grained predictions made by our analysis. We predict that parasupposition triggers should be ameliorated in the restrictor compared to the nuclear scope of *no;* and we predict that under *not* and *none*-type quantifiers alike, parasupposition triggers should be ameliorated if they appear within an embedded conjunction.

5.1 Prediction 1: restrictors versus nuclear scopes

Although the generalized quantifier *no* is semantically symmetric between its nominal and its verbal argument, embedding a parasupposition trigger in the nominal argument is acceptable, whereas this is unacceptable in the verbal argument. The contrast is illustrated in summarized in (60).

(58) a. #Aucun étudiant n' a mangé en cours, faisant des saletés.
no student NE has eaten in class, making some messes
'No student ate in class, making a mess.'
b. Aucun étudiant qui a mangé en cours, faisant des saletés,
no student who has eaten in class, making some messes,
n' est resté jusqu'au bout.
NE is stayed until the end
'No student who ate in class, making a mess, stayed until the end.'

(59) a. #Aucune crème pour la peau n'utilise de zirconite, provoquant des irritations cutanées.
no cream for the skin NE uses of zirconite, causing some irrations cutaneous
b. Aucune crème qui utilise de la zirconite, provoquant des irritations cutanées,
no cream that uses of the zirconite, causing some irritations cutaneous,
ne sera mise sur le marché
NE will-be put on the market
'No cream that uses zirconite, causing skin irritations, will put on sale.'

(60) If π is a parasuppositional expression, and N and V are nominal and verbal expressions respectively, the following acceptability patterns are found:

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[19] Two remarks should be added. First, if *S* is replaced with *CS* both for the assessment of the assertive condition and of the projective condition, the logic of the argument will remain unchanged. But if *S* is replaced with *CS* in the projective but not in the assertive condition, the problem discussed in the text will arise. Second, this problem might potentially be solved by replacing Non-eliminability with more gradient notions ("not enough informational value"), as argued on separate grounds by Spector 2025. Such gradient notions might also be useful to derive deviance under some *less than n*-type quantifiers if the contrast-based analysis sketched in Section 2.2.5 turns out to be insufficient.

a. #[No N] [V * π]
b. Ok [No [N * π]] V

This contrast is predicted by our account. The key is that, on standard analyses (e.g. Heim 1983, Schlenker 2009) , the projective condition yields a stronger inference in (60)b (namely *Every N G*) than in (60)a (namely *Every [N who Vs] G*). For this reason, Non-eliminability is satisfied in (60)b, as the parasuppositional expression does some work. This is laid out in greater detail in (61).[20]

(61) **Why [No [N * π]] V does not violate Non-eliminability**
a. Assertive condition: [No [N and π]] V
b. Projective condition: [Every N] π
c. Simpler alternative: [No N] V
d. Suppose that in world w, no N-individidual satisfies V and no N-individual satisfies π. Then the simpler alternative in c. is true in w, but the projective condition in b. is falsified in w.

Note that the strong projective condition in (61)a has been called into question on empirical grounds (e.g. Chemla 2009). The condition might be existential in force, thus: *[Some N] G*. This does not change the argument, as this inference too fails to be triggered by the simpler alternative in (61)c (and the situation in (61)d works just as well for the existential as for the universal version of the projective condition).

5.2 *Prediction 2: conjunctions under* none

One might initially think that parasupposition triggers are systematically deviant under *not* and *none* because these are negative expressions. But the facts are more complex: when a parasuppositional expression is embedded in a conjunction, it can appear in the scope of *none*-type quantifiers, as seen in (62)-(63). Here we replace *no NP* ('aucun…') with *there isn't a single NP* ('il n'y a pas un seul…') because the latter expression circumvents the need for the negative particle *ne*, whose repetition in conjunctions makes things independently difficult. (Note that in (62)c-(63)c, the participle could in principle be attached to the entire conjunct, but on semantic grounds this would make little sense, and it is thus safe to assume that the participle is attached to the second conjunct.)

(62) a. #Il n' y a pas un seul étudiant qui fume,
it NE there have not a single student who smokes,
prenant des risques énormes.
taking some risks enormous.
b. Il n' y a pas un seul étudiant qui à la fois fume,
it NE there have not a single student who a the time smokes,
prenant des risques énormes, et se soucie de sa santé.
taking some risks enormous, and SE worries of his health
'There isn't a single student that both smokes, taking enormous risks, and minds his health.'
c. Il n' y a pas un seul étudiant qui à la fois se soucie de sa santé
it NE there have not a single student who at the time SE minds of his health
et fume, prenant des risques énormes.
and smokes, taking some risks enormous
'There isn't a single student that both takes care of his health and smokes, taking enormous risks.'

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[20] Unsurprisingly, parasuppositionals expressions can be embedded in the nominal argument of *every,* which is also a downward-entailing environment. This is illustrated in (i) and schematized in (ii). Here too, the projective condition does some truth-conditional work by contributing a requirement that *[Every N] G*.

(i) Chaque étudiant qui fume, prenant des risques énormes, est inconséquent.
each student who smokes, taking some risks enormous, is inconsequential.
'Every student that smokes, taking enormous risks, is inconsequential.'

(ii) Ok [Every [N * G]] V

(63) a. #Il n' y a pas un seul étudiant qui mange des glucides,
it NE there have not a single student who eats some carbohydrates,
emmagasinant beaucoup de calories.
accumulating a-lot of calories
b. Il n' y a pas un seul étudiant qui à la fois mange des glucides
it NE there have not a single student who at the time eats some carbohydrates,
emmagasinant beaucoup de calories, et se soucie de sa santé.
accumulating a-lot of calories, and SE worries of his health.
'There isn't a single student that both eats carbohydrates, ingesting a lot of calories, and takes care of his health.'
c. Il n' y a pas un seul étudiant qui à la fois se soucie de sa santé
it NE there have not a single student who at the time SE worries of his health
et mange des glucides, emmagasinant beaucoup de calories.
and eats some carbohydrates, accumulating a-lot of calories
'There isn't a single student that both takes care of his health and eats carbohydratates, ingesting a lot of calories.'

The patterns we have seen so far are summarized in (64).

(64) a. #[No N] [V * π]
b. [No N] [[V * π] and V']
c. [No N] [V and [V' * π]]

They can be derived through a move that is standard in the literature on NPI-licensing (e.g. Chierchia 2004): (65)a,b trigger the implicature in (65)c. By contrast, the simpler alternative in (66)a triggers the implicature in (66)b. It is clear that (65)c is stronger than (66)b. As a consequence, if Non-eliminability takes into account implicatures, it has no reason to prohibit (65)a,b, as the parasupposition trigger does real truth-conditional work.

(65) a. [No N] [[V * π] and V']
b. [No N] [V and [V' * π]]
c. Implicature: [Some N] [V or [V' * π]]

(66) a. [No N] [V and V']
b. Implicature: [Some N] [V or V']

There is an alternative analysis, however. As we show in Appendix III, for (65)a, copied as (66)b, one gets a parasuppositional inference that is *not* made vacuous by the assertive component of the sentence. In a nutshell, this is because the parasupposition is triggered in the first conjunct and thus yields the strong inference in (67)b, which pertains to all N-individuals that satisfy *V*. The fact that there are no *N* individuals that satisfy both *V* and *V'* doesn't suffice to make the strong inference vacuous. Without further measures, however, this line of explanation does not extend to (65)b, copied as (68)a: the parasuppositional inference in (68)b is now made vacuous by the assertive component. As we explain in Appendix IV, however, there might be independent reasons to think that even in this case the parasupposition is strengthened to (68)c. Thus our targets might be derived even if we don't take implicatures into account.

(67) a. [No N] [[V * G] and V']
b. Parasupposition: [Every [N & V]] G

(68) a. [No N] [V and [V' * G]]
b. Parasupposition: [Every [N & [V and V']] G
c. Strengthened parasupposition: [Every [N & V]] G

We further discuss in Appendix III the case of disjunctions embedded under *none*-type operators. We show that on either theory considered here to derive the behavior of embedded conjunctions, disjunctions are predicted *not* to obviate the prohibition again parasupposition triggers under *none*, and we argue that the prediction is correct.

6 Flip-flop Effects and the Connection with Positive Polarity Items

Since monotonicity considerations play a crucial role in the licensing of parasuppositions, we expect that parasuppositions should give rise to "flip-flop" effects, in the sense that stacking negative expressions should make parasuppositions acceptable or unacceptable depending on the logical properties of their environments. We derive the prediction, display examples that bear it out, and suggest that other examples probably don't, which might call for a 'localized' version of Non-eliminability.

6.1 Predicting flip-flop effects

To see that flip-flop effects are expected on the basis of Non-eliminability, consider once again our explanation of the deviance of a parasuppositional expression immediately embedded under *not*. The key was that *not [p, g]* makes the two contributions in (69): (69)a is assertive, while (69)b, equivalent to (69)b', is projective; and their conjunction is equivalent to the contribution made by the simpler alternative *not p* in (70).

(69) Two contributions of *not [p, π]*
a. Assertive contribution: S |= not [p and π]
b. Projection contribution: If s' is the local context of π relative to S, s' |= π
b'. S |= p => π

(70) Contribution of *not p*
S |= not p

Now let us add one negation to each alternative, as in (71).

(71) Two contributions of *not not [p, π]*
a. Assertive contribution: S |= not not [p and π]
a'. S |= p and π
b. Projective contribution: S |= p => π

After negation elimination, the assertive contribution in (71)a boils down to the assertion of *p and π*. And because negation does not affect local context computation and therefore parasupposition projection, the projective condition in (71)b-b' is identical to that in (69)b-b'. It is immediate that *not not [p, π]* contributes the information *[p and π],* while the simpler alternative *not not p* just contributes the information *p*. Therefore Non-eliminability should be satisfied when we add a negation to *not [p, π]*. In other words, there should be amelioration upon addition of a second negative expression.

(72) Contribution of *not not p*
S |= p

We note that the same logic applies if the highest negation is replaced with *never* (or a synonym), as shown in (73).

(73) Two contributions of *never not [p, π]*
a. Assertive contribution: S |= never not [p and π]
a'. S |= always not not [p and π]
a". S |= always [p and π]
b. Projective contribution: S |= always [p => π]

(74) Contribution of *never not p*
S |= always p

6.2 Detecting flip-flop effects

We already provided above examples of sentences that are acceptable with parasupposition triggers, but become degraded when embedded under *not* or *none*-type operators. We leave a more systematic investigation of this type of flip-flop effect for future research, and turn to the case in which degraded sentences get ameliorated upon embedding.

We find some data subtle and can only state confidently that there are *some* cases of amelioration upon embedding under a further negative expression, as stated in (75)

(75) Some APPs that are degraded under negation can be ameliorated upon embedding under an additional negative expression.

The prediction is borne out in (76). Owing to the difficulty of stacking two negative expressions immediately on top of each other (as is: *never… not* ['jamais… pas']), we use in (76)c the expression *there isn't a single meeting in which* ('il n'y a pas une seule réunion où'), which makes embedding relatively smooth. Throughout, note that there is in principle an attachment ambiguity for the participle, but that only the low (target) attachment makes sense.

(76) a. Anne a mangé, emmagasinant beaucoup de calories.
Anne has eaten, acquiring a-lot of calories.
'Anne has eaten, thus ingesting a lot of calories.'
b. #Anne n' a pas mangé, emmagasinant beaucoup de calories.
Anne NE has NOT eaten, acquiring a-lot of calories.
c. Il n' y a pas eu une seule réunion où Anne
it NE there have not had a single meeting where Anne
n' a pas mangé, emmagasinant beaucoup de calories.
NE has not eaten, acquiring a-lot of calories.
'There isn't a single meeting inwhich Anne didn't eat, thus ingesting a lot of calories.'

Importantly, the APP could can meaningfully modify the low VP *eat*. The only higher VP is *(there) is (a single meeting),* and it would make no sense for this construction to be modified by the APP (e.g. there would be no subject corresponding to Anne). The example in (77) makes the same point.

(77) a. Anne a chanté, incommodant ses voisins.
Anne has drunk, disturbing her neighbors.
'Anne has sung, thus inconveniencing her neighbors.'
b. #Anne n' a pas chanté, incommodant ses voisins.
Anne NE has not sung, disturbing her neighbors.
c. Il n' y a pas eu un seul jour où Anne
it NE there have not had a single day where Anne
n' a pas chanté, incommodant ses voisins.
NE has not sung disturbing her neighbors.
'There is a single day when Anne didn't sing, thus inconveniencing her neighbors.'

The same argument can be made with embedding under the negative expression *without eating/singing*—which can be paraphrased, to bring out the negative nature of *without*, as: *cooccurring with no event of eating/singing*. The target sentences appear in (78)-(79). Importantly, considerations of plausibility force the modifier headed by *after/without* to be attached to the embedded verb.

(78) a. (?) Anne travaille après avoir [mangé, emmagasinant beaucoup de calories].
Anne works after having eaten, acquiring a-lot of calories.
'Anne works after eating, thus ingesting a lot of calories.'
b. #Anne travaille sans avoir [mangé, emmagasinant beaucoup de calories].
Anne works without having eaten, acquiring a-lot of calories.
c. Anne ne travaille jamais sans avoir [mangé, emmagasinant beaucoup de calories].
Anne NE works never without having eaten, acquiring a-lot of calories.
'Anne never works without having eaten, thus ingesting a lot of calories.'

(79) a. Anne médite après avoir [chanté, incommodant ses voisins].
Anne médite after having sung, disturbing her neighbors.
'Anne meditates after singing, thus inconveniencing her neighbors.'
b. #Anne médite sans avoir [chanté, incommodant ses voisins].

Anne meditates without having sung, disturbing her neighbors.
c. Anne ne médite jamais sans avoir [chanté, incommodant ses voisins].
Anne NE meditates never without having sung, disturbing her neighbors.
'Anne never meditates without having sung, thus inconveniencing her neighbors.'

We conclude that a global computation of Non-eliminability is *sometimes* necessary, as predicted.

6.3 From global to local conditions?

We do not think this (weak) claim exhausts the data, however. There are cases in which double negative embedding doesn't restore the acceptability of a clause containing a parasupposition trigger, as illustrated in (80)b-(81)b, which are not clearly better than (80)a-(81)a.

(80) a. #Anne n' a pas mangé, emmagasinant beaucoup de calories.
Anne NE has NOT eaten, acquiring a-lot of calories.
b. #Il n' est pas vrai qu' Anne n' a pas mangé, emmagasinant beaucoup de
Il NE est not true that Anne NE has NOT eaten, acquiring a-lot of
calories.
calories.

(81) a. #Anne n' a pas chanté, incommodant ses voisins.
Anne NE has not sung, disturbing her neighbors.
b. #Il n' est pas vrai qu' Anne n' a pas chanté, incommodant ses voisins.
Il NE est not true that Anne NE has not sung, disturbing her neighbors.

One solution, tailored to this specific case, is to posit that (80)b-(81)b have an echoic use, with the inference that someone asserted the sentences in (80)a-(81)a. Since the latter are deviant, the echoic reading itself should be. Another solution, which might have broader applicability, is to posit that, under conditions to be determined, a localized version of Non-eliminability must be invoked alongside the global version. This would dovetail with important debates in formal pragmatics, pertaining to operations that are intuitively global but turn out to have local incarnations as well (implicatures are a prime example, see for instance Chierchia et al. 2012). We leave this question for future research.

6.4 The connection with PPIs and Homer's theory of polarity

Flip-flop effects (when they arise) make it possible to uncover a further restriction on parasupposition triggers, stated in (82), which makes them even more similar to PPIs. As we will see, this restriction follows from Homer's (2011, 2021) theory of polarity.

(82) **Cooccurrence with NPIs and PPIs**
Under negative expressions, parasupposition triggers can co-occur with PPIs but not with NPIs contained in the clause they modify.

For a PPI and an NPI to have a chance to be simultaneously acceptable, we must stack two negative operators on top of each other. As can be seen in (83), both the NPI *quoi que ce soit* ('anything') and the PPI *quelque chose* ('something') are acceptable under two negative expressions, but when a participle is added, only the PPI is acceptable, as shown in (84)b.[21]

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[21] Two remarks should be added. First, we use the APP 'thus contravening to orders' rather than 'thus ingesting a lot of calories' because the minimizer component of the NPI might independently make 'a lot' degraded. Second, we find the target contrasts less clear in *without* constructions because the baselines with NPIs seem to us to be a bit degraded, as in (i) below.

(i) Anne ne travaille jamais sans avoir mangé ??quoi que ce soit / quelque chose.
Anne NE works never without having eaten ??what that it be / some thing.

(83) Il n' y a pas eu une seule réunion où Anne
it NE there have not had a single meeting where Anne
n' a pas mangé quoi que ce soit / quelque chose.
NE has not eaten, what that it be / some thing.
'There isn't a single meeting in which Anne didn't eat something / anything.'

(84) a. Il n' y a pas eu une seule réunion où Anne
it NE there have not had a single meeting where Anne
n' a pas mangé, contrevenant aux ordres.
NE has not eaten, contravening to-the orders.
'There isn't a single meeting in which Anne didn't eat, thus contravening to orders.'
b. Il n' y a pas eu une seule réunion où Anne
it NE there have not had a single meeting where Anne
n' a pas mangé #quoi que ce soit / quelque chose,
NE has not eaten what that it be / some thing.
contrevenant aux ordres.
contravening to-the orders.
'There isn't a single meeting in which Anne didn't eat #anything / something, thus contravening to orders.'

This pattern only highlights the similarity between parasupposition triggers and PPIs. The reason is that the above correlation follows from generalizations uncovered in Homer 2011, 2021 about the co-occurrence of NPIs and PPIs. In a nutshell, Homer proposes that for an NPI or PPI to be licensed, one should find *some* sufficiently large "eligible domain" in a sentence which satisfies its logical specifications, and thus creates a downward-entailing environment (for NPIs) or a non-downward-entailing one (for PPIs). By an "eligible domain", we mean (as a first approximation) a constituent that includes the projection for negation, as argued at length by Homer (2011, 2021). But this is not enough. As Homer shows, this constituent should simultaneously satisfy the requirements of all the polarity items it contains, checked cyclically (i.e. inside-out). This predicts intricate patterns of acceptability, illustrated in (85).

(85) a. It is impossible that <u>someone stole something</u>.
b. **It is impossible that anyone stole anything**.
c. **It is impossible that anyone <u>stole something</u>**.
d. *It is impossible that **someone stole anything**.
(Homer 2011)

In (85)a, the (underlined) embedded clause in isolation creates upward-entailing environments for the two PPIs it contains. In (85)b, the entire sentence (boldfaced) creates downward-entailing environments for the two NPIs it contains. In (85)c, in a first cyclic step, the embedded VP (underlined) taken in isolation creates an upward-entailing environment for the PPI *something*. In a second cyclic step, the entire sentence (boldfaced) creates a downward-entailing environment for the NPI *anyone*. Remarkably, in (85)d this cyclic procedure fails. To satisfy the requirements of *someone*, one must consider the embedded clause (boldfaced) in isolation. But it contains the NPI *anything,* which is unlicensed and could not have been licensed in an earlier step.

Descriptively, the examples in (83)-(84) follow from Homer's theory on the assumption that parasupposition triggers are PPIs. Our examples have the structure given in (86).

(86) $[_{\text{NegP_2}}$NEGATIVE … $[_{\text{NegP_1}}$not Ann [[eat something / #anything] [contravening to orders]]]]

The Homer-domains to consider are NegP$_1$, which contains the embedded negation, and NegP$_2$, which in addition contains the matrix negative expression. For *anything* to be licensed, we must consider NegP$_1$, which creates a downward-entailing environment for it. But in NegP$_1$, the parasupposition trigger is not in an appropriate (non-downward-entailing) environment, and furthermore smaller

'Anne never works without having eaten anything / something.'

constituents are not eligible according to Homer (they don't contain a projection for negation). No problem arises for the PPI *something*, by contrast: relative to $NegP_2$, both it and the parasupposition trigger appear in non-downward-entailing environments.

One last remark is that in other environments than those discussed above, APPs can sometimes modify VPs containing NPIs, as is illustrated in (87) (which echoes (36) above).

(87) [Si Anne [$_{NegP\_1}$ [mangeait **quoi que ce soit**], [contrevenant aux ordres]], je la critiquerais].
if Anne ate what that it be, contravening to-the orders, I her would-criticize.
'If Anne ate anything, thus contravening to orders, I would criticize here.'

As far as we can tell, this particular pattern follows from Homer's theory on the (uncontroversial) assumption that a conditional is an eligible domain for the licensing of NPIs. The analysis goes as follows. First, the appositive participle *qua* PPI is licensed in $NegP_1$, which is an upward-monotonic environment. Second, the boldfaced NPI is licensed in the larger eligible domain defined by the entire sentence. Relative to that larger domain, the NPI is in downward-monotonic environment, as desired. The boldfaced PPIs isn't, but that is fine, as it was licensed in an earlier cycle.[22]

We leave for future research a full exploration of the PPI behavior of parasupposition triggers—as well as their connection to Homer's theory. This PPI-like behavior could suggest that parasupposition triggers are *just* PPIs, and that there is no need for our manner-based derivation of their deviance under negation and *none*-type operators. But this alternative analysis will likely fall short on two levels. On an empirical level, we saw repeatedly that parasupposition triggers under negation and *none*-type operators can be ameliorated if the parasupposition enters in some discourse relation. This suggests that a violation of manner is indeed the cause of the baseline deviance. On a theoretical level, saying that parasupposition triggers are "just" PPIs is less explanatory than deriving their distribution from well-established pragmatic principles. In fact, precisely this was attempted by Solt and Wilson 2021 and Nicolae and Solt 2025, as discussed in Section 4.3. In the end, it might be that parasupposition triggers are *also* (rather than *just*) PPIs.

7 Parenthetical Conjuncts as Parasupposition Triggers

Appositive participles are not the only parasupposition triggers. In this section and the next, we propose that two further constructions belong to the family: parenthetical conjuncts, and (as foreshadowed in the introduction) post-speech gestures. In both cases, there are slight differences relative to participles that clearly call for further investigation.

7.1 Parenthetical conjuncts

Parenthetical conjuncts are illustrated in (88), where they give rise to clear projection patterns.

(88) a. Ann will travel (and take enormous risks).
b. Maybe Ann will travel (and take enormous risks).
=> if Ann travels, will take enormous risks

[22] This is not the last word on this construction. Homer 2011 (p. 92) discusses difficulties raised by conditionals, as in (i). The problem is that the pattern in (85)d, copied in (ii), is not clearly replicated in (i)c.

(i) a. If someone stole a camera, we're in trouble.
b. If John stole anything, we're in trouble.
c. ?[If **someone [$_{NegP\_1}$ stole anything]**, we're in trouble].
(Homer 2011)

(ii) *It is impossible that **someone stole anything**. (Homer 2011)

In (ic), the NPI cannot be licensed in $NegP_1$. It must be licensed relative to something at least as large as the *if*-clause, but the latter is a downward-entailing environment, which according to Homer's theory should anti-license the PPI *someone*.

c. Will Ann travel (and take enormous risks)?
=> if Ann travel, she will take enormous risks

Since parentheses are not pronounced as such, we take parenthetical conjuncts to be most clearly realized by having a pause before *and*, by downstressing the entire parenthetical, and by producing it with a co-speech gesture resembling this one: .[23]

7.2 Key properties

7.2.1 Projection patterns

In French, parenthetical conjuncts pattern with participles in the projection patterns they give rise to, as illustrated in (89).

(89) a. Est-ce qu' Anne voyage (et prend des risques énormes).
is it that Anne travels (and takes some risks enormous).
'Does Anne travel (and take enormous risks)?'
=> if Anne travels, she takes enormous risks
b. Peut-être qu' Anne voyage (et prend des risques énormes).
maybe that Anne travels (and takes some risks enormous).
'Maybe Anne travels (and takes enormous risks.'
=> if Anne travels, she takes enormous risks
c. Si Anne voyage (et prend des risques énormes), je ne serai pas content.
if Anne travels (and takes some risks enormous), I NE will-be not happy.
'If Anne travels (and takes enormous risks), I won't be happy.'
=>? if Anne travels, she will take enormous risks
d. Si Anne devient correspondante à Damas, est-ce qu'elle va voyager
if Anne becomes correspondent in Damascus, is it that she will travel
(et prendre des risques énormes)?
(and take some risks enormous)?
'If Anne becomes a correspondent in Damascus, will she travel (and take enormous risks)?'
=> if Anne becomes a correspondent in Damascus, if she travels, she will take enormous risks
e. Peut-être que chaque étudiant voyage (et prend des risques énormes).
maybe that each student travels (and takes some risks enormous).
'Maybe each student will travel (and take enormous risks).'
=> for every student x, if x travels, x takes enormous risks.

Still, we take the inferential data with parenthetical conjuncts to be a bit less sharp than those with appositive participles. We believe this is due to an ambiguity of the parenthetical, which allows for a corrective reading.[24] On the corrective reading, (89)d means in essence: *If Ann travels (or rather if she travels and takes enormous risks), I won't be happy*. Our impression is that this reading can be excluded by clearly downstressing the parenthetical conjunct, and marking the left parenthesis with the co-speech gesture , already mentioned above.

7.2.2 Narrow scope interpretation

In view of general properties of conjuncts (which function as syntactic units), we expect that parenthetical conjuncts should display a narrow scope behavior. This is borne out: the parenthetical

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[23] https://www.gettyimages.fr/detail/illustration/pssst-illustration-libre-de-droits/1148015083?adppopup=true
[24] Corrective readings of parentheticals/appositives can arguably be found with modifiers rather than conjuncts, as in (i) below (see for instance AnderBois et al. 2015 and Schlenker 2021 for discussion).

(i) If a professor, a famous one, publishes a book, he will make a lot of money. (Wang et al. 2005, cited in Nouwen 2014) conjunct in (90)b differs from the clausal parenthetical in (90)a in being interpreted in the scope of the attitude verb.

(90) *Context:* Anne's son has a life-threatening disease.
a. Anne a peur que son fils prenne du Zopaclon (cela augmentera ses chances de survie).
Anne has fear that her son take of Zopaclon (that will-incrase his chances of survival).
'Anne fears her son is taking Zopaclon (this will increase his survival odds).'
≠> Anne fears her son might increase his survival odds
b. Anne a peur que son fils prenne du Zopaclon (et augmente ses chances de survie).
Anne has fear that her son take of Zopaclon (and increase his chances of survival).
'Anne fears her son is taking Zopaclon (and increasing his survival odds).'
=> Anne fears her son might increase his survival chances (hence pragmatic oddity)

The same contrasts are found in (91).

(91) *Context:* Anne's son is sick.
a. Anne sait que son fils prend du Zopaclon (cela met sa vie en danger).
Anne knows that her son takes of Zopaclon (that puts his life at risk).
'Anne knows that her son is taking Zopaclon (this puts his life at risk).'
≠> Anne knows her son is putting his life at risk
b. Anne sait que son fils prend du Zopaclon (et met sa vie en danger).
Anne knows that her son takes of Zopaclon (that puts his life at risk).
'Anne knows that her son is taking Zopaclon (and putting his life at risk).'
=> Anne knows her son is putting his life at risk

7.2.3 *Informativity*

Parenthetical conjuncts are subject to clear informativity requirements. (92)a is degraded. (92)b is acceptable but has an irrelevant reading, namely: *Anne traveled to Syria and thereafter moved around* (presumably in Syria).

(92) Anne a voyagé en Syrie
Anne has traveled to Syria
'Anne traveled to Syria
a. #(et a voyagé).
(and has traveled).
b. (et s' est déplacée).
(and SE is moved).
(and moved around).

Similarly, (93) is sharply degraded.

(93) *Context:* Ann is known to have a father and a mother.
#La semaine dernière, Anne a invité ses parents (et invité son père).
the week last Anne has invited her parents (and invited her father).

7.2.4 *Deviance under negative expressions*

As was the case for APPs, parenthetical conjuncts are degraded under *not* and *none*, although the contrasts may be more subtle (they are clearly context-dependent, owing to the possibility of obviation through discourse relevance).

(94) a. <#> Anne n' a pas voyagé (et pris des risques énormes).
Anne NE has not traveled (and taken some risks enormous).
b. #Aucun étudiant n' a voyagé (et pris des risques énormes).
no student NE has *traveled (and taken some risks enormous).*
a'. Anne a voyagé (et pris des risques énormes).
Anne has traveled (and taken some risks enormous).
'Anne has traveled (and takend enormous risks).'

b'. Certains étudiants ont voyagé (et pris des risques énormes).
some students have traveled (and taken some risks enormous).
'Some students have traveled (and taken enormous risks).'

(95) Talking about a lottery:
a. #Anne n' a pas trouvé les bons numéros (et gagné 1000€).
Anne NE has not found the good numbers (and won 1000€).
b. #Aucun joueur n' a trouvé les bons numéros (et gagné 1000€).
no player NE has found the good numbers (and won 1000€).
a'. Anne a trouvé les bons numéros (et gagné 1000€).
Anne has found the good numbers (and won 1000€).
'Anne found the winning numbers, thus winning €1000.'
b'. Plusieurs joueurs ont trouvé les bons numéros (et gagné 1000€).
several players have found the good number (and won 1000€).
'Several players found the winning numbers, thus winning €1000.'

7.2.5 *Rescuing upon further embedding*

It is worth noting that amelioration can be obtained upon embedding under a further negative expression, in line with the findings of Section 6.2. Thus (96) and (97) are more acceptable than (94)a and (95)a respectively.

(96) Il n' y pas eu un seul mois où Anne n' a pas
itNE there not had a single month where Anne NE has not
voyagé (et pris des risques énormes).
traveled (and taken some risks enormous).
'There wasn't a single month when Ann didn't travele (and take enormous risks).'

(97) Talking about a lottery that takes place every month:
Il n' y pas eu un seul mois où Anne n' a pas
itNE there not had a single month where Anne NE has not
trouvé les bons numéros (et gagné 1000€).
found the good numbers (and won 1000€).

7.2.6 *Obviation through relevance*

As was the case for APPs, parenthetical conjuncts become acceptable under *not* and *none* when they serve to explain the quantified statement.[25] As in other cases of obviation, intonation probably plays a role as well, an issue we leave for future research.

(98) Uttered during the Covid pandemic:
a. Mon étudiant ne va pas rendre visite à sa famille (et mettre en danger
my student NE will NOT pay visit to his family (and put in danger
ses parents).
his parents).
'My student won't visit his family (and put his parents at risk).'
b. Aucun de mes étudiants ne va rendre visite à sa famille
none of my students NE will pay visit to his family
(et mettre en danger ses parents).
(and put in danger his parents).
'My student won't visit his family (and put his parents at risk).'

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[25] In our French examples, we rely on constructions involving the complex VP *[visit his family and thus put this parents at risk]* rather than *[visit his parents and thus put them at risk]* because the latter is hard to embed under the past tense auxiliary *have*. The reason is that the clitic for them, 'les', needs to cliticize to the auxiliary, but since the clitic starts out in the second conjunct (with the structure *have [visit his parents and thus put them at risk]*), this is impossible. The construction we use instead obviates the need for anaphora and cliticization.

7.3 Parasuppositions versus informational requirements on downstressing

Since our parenthetical examples involve downstressing, it is natural to analyze them in terms of the semantic correlate of downstressing, analyzed in Rooth 1992. In (99)a, analyzed as in (99)b, Rooth's initial requirement is that the value of the antecedent clause (underlined) should be part of the focus value of the target clause (underlined and downstressed). This focus value is obtained by replacing 7 with its alternatives in the computation of the value of the boxed element.

(99) a. 5 is less than or equal to 5 and (of course) 7_F is less than or equal to itself (as well).
b. [5 is less than or equal to 5]$_1$, and (of course) [7_F λi t_i is less than or equal to itself]~1.
(Rooth 1992)

To handle example like (100)a, one must refine the analysis to allow downstressing to be justified by entailment. For present purposes, we can take the requirement to be that the antecedent should entail an element of the focus value of the target. This requirement is satisfied in (100)a in case it is presupposed that if Mary calls me a Republican, she insults me. The requirement is not easily satisfied in (100)b because it is hard to presuppose that if Mary insulted me, then she called me a Republican.

(100) a. First Mary called me a Republican and then $John_F$ insulted me.
b. <#> First Mary insulted me and then $John_F$ called me a Republican.
(modified from Rooth 1992)

Rooth 1992 applies his analysis to cases without focus, as in (101)a. Here the focus semantic value of the target is identical to its ordinary value, and the requirement that a member of the focus value be entailed by the antecedent boils down to the requirement that the target be entailed by the antecedent; the reason is that the focus value just contains the ordinary value in this case. Taking into account the tweak in terms of entailment, the requirement in (101)b is just that studying semantics should entail studying linguistics. (If *semantics* and *linguistics* are reversed, as in (101)c, the result is acceptable only to the extent that *studying linguistics* is taken to contextually entail *studying semantics*—for instance, if the only linguistics class offered is in semantics).

(101) a. We are supposed to take statistics and [study semantics]$_1$ this term, but I don't like [studying semantics]~1. (Modified from Rooth 1992).
b. We are supposed to take statistics and [study semantics]$_1$ this term, but I don't like [studying linguistics]~1.
c. We are supposed to take statistics and [study linguistics]$_1$ this term, but I don't like [studying semantics]~1.

It is clear from (100)a that the entailment relation that holds between the antecedent and a member of the focus value of the target can easily be accommodated. This, in turn, could give a very natural account of parasuppositions triggered by parenthetical/downstressed material, as illustrated in (102).

(102) a. Maybe Ann will travel (and take enormous risks).
b. Maybe Ann will [travel]$_1$ (and take enormous risks~1).

Using Rooth's ideas, there must be a presupposition that the antecedent *travel* entails the downstressed element *take enormous risks*. The most natural implementation of this idea is to require that entailment should hold relative to the local context of *travel* (which incorporates information about the subject *Ann*, see e.g. Schlenker 2009). If we call this local context c', we have the requirement in (103).

(103) c' |= travel => take enormous risks

In (102), there is one additional constraint, which is a special case of the requirement that an element should not be trivial in its local context (e.g. Stalnaker 1978, Schlenker 2009): the (downstressed) second conjunct should not be trivial relative to its local context. The local context of the second conjunct is obtained by intersecting the local context of the first conjunct with the value of that conjunct, hence the requirement in (104)a, which is equivalent to (104)b.

(104) a. c' ∩ **travel** |≠ take enormous risks
b. c' |≠ travel => take enormous risks

(104)b directly contradicts (103). Applying global accommodation presumably won't help, because it would have the effect of simultaneously affecting the value of *c*' in (103) and (104), hence leaving the contradiction unchanged. On the other hand, no contradiction arises if the requirement in (103) (but not that in (104)) is evaluated relative to the speaker's local context *s*', i.e. the local context of the first conjunct obtained from the speaker's belief set rather than from the context set:

(105) s' |= travel => take enormous risks

In sum, a special case of Rooth's theory of downstressing derives the projection behavior of downstressed parentheticals, but it must still be stipulated that the condition is evaluated relative to the speaker's local context of the downstressed element rather than to the standard local context. Explaining why this is a possibility is part of the broader question of explaining why parasuppositions can project like presuppositions while constraining something other than the context set (namely the speaker's beliefs, in the simplest version of our analysis).

7.4 Intermediate summary

In sum, parenthetical conjuncts (disambiguated with an appropriate co-speech gesture) share the main target properties of APPs: they give rise to a projection behavior reminiscent of presuppositions, but unlike presuppositions, they are preferably non-redundant; they are usually deviant under negation and *none*-type operators, but they can ameliorated upon further negative embedding, or when the projected inference helps justify the main claim. Parenthetical conjuncts can thus be analyzed as parasupposition triggers.

8 Post-speech Gestures as Parasupposition Triggers

We now come back to post-speech gestures, to buttress the claim that they too can fruitfully be analyzed as parasupposition triggers.

8.1 Key properties

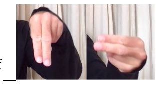
We already saw in Section 1.2 that post-speech gestures modifying VPs in English share key properties of APPs: they give rise to projection patterns similar to presuppositions (see (8) above), they are subject to an informativity requirement (see (9)), and they are degraded under *not* and *none*, a point amply discussed in the literature, and illustrated above (see (10)). In our judgment, the same facts extend to French examples with post-speech gestures. For brevity, we focus on additional properties of interest in this piece.

8.1.1 Narrow scope interpretation

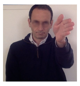
We argued in Section 1.2 against the "wide scope only" analysis of post-speech gestures. In brief, their distribution and interpretation forces the "wide scope only" theory to posit powerful mechanisms of modal subordination—so powerful that it becomes mysterious why post-speech gestures are degraded under negative expressions. But can we find more direct evidence for narrow scope interpretation?

In the following examples, a wide scope interpretation seems very difficult. In (106)a, the gesture representing the model falling (into the pool) is interpreted within the scope of the attitude verb. By contrast, this interpretation is not obligatory with the ARC in (106)b, as expected in view of the wide scope behavior of ARCs. In (106)c, a feeling of contradiction is obtained if a conjunct within the embedded clause contradicts the content of the post-speech gesture, which again suggests that the latter must be interpreted in the scope of the attitude verb. Similar facts hold in (107)b.

(106) *Context:* A photo shoot has been organized right next to a swimming pool.
Le photographe voudrait que le modèle fasse un pas de plus en arrière
the photographer would-like that the model do-subj one step of more in back
'The photographer wants the model to take one further step backwards

a. – FALL^{plouf}_.
– FALL^{plouf}.
– FALL^{plouf}.'
=> the photographer wants the model to fall into the pool
b. , ce qui le fera tomber dans la piscine.
, it that him will-make fall in the swimming-pool.
, which will make him fall into the swimming pool.'
≠> the photographer wants the model to fall into the pool
c. # – FALL^{plouf}, mais en restant bien droit.
– FALL^{plouf}, but in staying well upright.

(107) Anne pense qu' Astérix va punir son ennemi
Anne thinks that Asterix will punish his enemy
'Ann thinks that Asterix will punish his enemy

a. – SLAP_.
– SLAP.
– SLAP.'
=> Ann thinks that Asterix will physically punish his enemy
b. ce qu'il fera en le frappant.
it that he will-do in him hitting.
which he will do by punching him.'
≠> Ann thinks Asterix will physically punish his enemy
c. # – SLAP, mais seulement par des paroles.
– SLAP, *but only with some words.*

We conclude that these direct tests dovetail with the indirect argument developed in Section 1.2: in our target sentences, post-speech gestures are preferably interpreted with narrow scope. Whether they can also have a less salient wide scope interpretation is a question that would require a more detailed investigation.

8.1.2 Deviance under not *and* none: *rescuing by further embedding*

While the deviance of post-speech gestures under *not* and *none* has been discussed in the literature (e.g. Schlenker 2018a,b), rescuing upon embedding under further negative operators has not been discussed yet. It is clearly found in (108)b, which contrasts with (108)a. While the point of attachment of sentence-final post-speech gestures is sometimes ambiguous, this is not the case here, as the post-speech gesture is embedded with the first clause of a disjunction. The same logic is at work in (109).

(108) *Context:* Anne's three-year old loves going down the slide every afternoon, but he has difficulty climbing on it.
–Est-ce que le fils d' Anne arrive à monter sur le tobogan tout seul?
is it that the son of Anne manges to go-up on the slide all alone?
'Does Anne's son manage to climb the slide on his own?'

a. –Oui - <#?> Elle ne l' a pas aidé – UP_.
–Yes - she NE him has not helped – UP.
b. –Non. Il n' y a pas eu un seul après-midi où

–no. there NE there have not had a single afternoon where elle ne l' a pas aidé – UP ou carrément porté.
she NE him has not helped – UP or outright carried.
'-No. There a single afternoon when she didn't help him – UP or outright carry him.'

(109) *Context:* In a new comic, Asterix is faced with a Roman enemy who seeks to assassinate him on every page. A hasn't read the comic but B has just finished doing so.
–A: Est-ce qu' Astérix a puni son ennemi?
–A: is it that Asterix has punished his enemi?
'-A: Has Asterix punished his enemy?'
a. #–B: Non, il ne l' a pas puni – SLAP.
–B: no, he NE him has not punished – SLAP.
b. –B: Oui, il n' y pas une seule page où il ne l'a pas puni – SLAP ou carrément mis KO.
–B: oui, it NE there not a single page where punished – SLAP or outright put KO.
'B: Yes, there isnt a single page on which he didn't punish him – SLAP or outright knock him out.'

8.1.3 Deviance under not *and* none*: obviation through discourse relevance*

It should be added that sometimes post-speech gestures under *not* or *none* can be ameliorated if they provide a reason why certain actions didn't happen, which mirrors the behavior of APPs and parenthetical conjuncts. We note again that intonation might play an important role, and thus the examples below are not entirely controlled. They just suggest that deviance under *not* and *none* can be ameliorated when the gesture enters in discourse relations.

(110) *Context:* A photo shoot has been organized right next to a swimming pool.
a. Le modèle est tout au bord de la piscine, mais il ne va pas
the model is all at-the edge of the swimming-pool, but he NE will not

faire un pas de plus en arrière – FALL^{plouf}.
make one step of more in back – FALL^{plouf}.
'The model is at the edge of the swimming pool, but he won't take an additional step backwards – FALL^{plouf}.'
=> the reason the model won't take an additional step backwards is that he would fall into the swimming pool
Video: https://www.dropbox.com/scl/fi/09z5qx9wj1hkosyx0ol7s/Model-won-t-plouf-trimmed_processed.mov?rlkey=l6906mvp9zmsj472brmzm3rwn&dl=0
b. Tous les modèles sont tout au bord de la piscine, mais
all the models are right at-the edge of the swimming-pool, but
aucun d'entre eux ne va faire un pas de plus en arrière – FALL^{plouf}.
none of among them NE will make one step of more in back – FALL^{plouf}.
'All the models are at the edge of the swimminig pool, but none of them will take an additional step backwards – FALL^{plouf}.'
=> the reason the models won't take an additional step backwards is that they would fall into the swimming pool
Video: https://www.dropbox.com/scl/fi/wd687plpt47zes4nhef3t/Models-will-none-trimmed_processed.mov?rlkey=pl9p9jt5m95z8liyv7qi2hth2&dl=0

8.2 Intermediate summary

Post-speech gestures have been analyzed as displaying the pattern of ARCs and in triggering supplements, in particular because of their deviance under negative expressions. As we argued at the outset, their apparent embeddability forces supplement-based theories to posit powerful mechanisms of modal subordination that in the end make deviance in negative environments somewhat mysterious. Here we provided a direct argument that post-speech gestures display all the hallmarks of parasuppositions: they display narrow scope interpretation while giving rise to projection, they are subject to an informativity constraint, and they are often degraded under negative expressions—but like other parasupposition triggers, they can be ameliorated when negative expressions are stacked, or when the post-speech gesture enters in certain discourse relations.

This parasuppositional theory thus offers a strong alternative to the supplemental analysis of post-speech gestures. But this is only the beginning of a systematic study. In particular, it would be important to test more systematically the behavior of post-speech gestures under other downward-monotonic expressions, in particular in comparison with appositive participles.[26] The scopal properties of post-speech gestures should also be compared more systematically to those of ARCs and APPs. We leave these questions for future research.

9 Conclusion

To conclude, parasuppositions have three key properties. First, they project like presuppositions (P1). Second, they are typically informative, contrary to presuppositions (P2). Third, parasupposition triggers are usually degraded under negative expressions, but they can be ameliorated upon further negative embedding or when the parasuppositional inference plays an independent discourse role (P3). We offered a theory on which parasuppositions must be non-trivial relative to their standard local context (hence P2), but trivial relative to their speaker's local context, which is more specific (hence P1). Under a single negation or *none*-type operator, they can be eliminated without affecting the truth conditions, which yields a violation of manner. It can be avoided if the local environment is made non-negative by stacking further negative expressions, or if the parasuppositional inference enters in some discourse relations (this P3). We motivated our analysis by a detailed investigation of the behavior of appositive present participles in French, and argued that further constructions, such as parenthetical conjuncts and post-speech gestures, display a parasuppositional behavior as well.

Several questions should be explored in future research. On an empirical level, more work is needed on multiple aspects of our target constructions.[27] One should also look for further constructions that display the hallmarks of parasuppositions, both in French and across languages.[28] On a theoretical

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[26] Some examples were discussed in the literature, such as (i), from Schlenker 2018a. Importantly, deviance under *it's unlikely that* does not follow from the present theory.

(1) a. It's likely/#It's unlikely that John helped his son, which (by the way) he did by lifting him.
b. It's likely/#It's unlikely that John helped his son – UP.

[27] In particular: 1. Restrictions under *less than n*-type quantifiers and further downward-monotonic operators besides negative ones must be better understood. 2. The detailed syntax of APPs needs to be investigated, notably to determine to which constituents it can attach to. 3. Marginal cases of wide scope of behavior of APPs must be revisited. 4. The connection with narrow scope ARCs must be explored. 5. The role of discourse relations will have to be investigated in greater detail. 6. The fine-grained predictions of Section 5 were tested with APPs but not with parenthetical conjuncts and post-speech gestures; this will have to be done. 7. An important open problem appears in (i): acceptable cases of *less than n*-type quantifiers tend to come with a non-emptiness requirement when they are modified by parasuppositional expressions (this requirement disappears if the APPs is removed).

(i) Moins de 5 joueurs vont trouver les bons numéros, gagnant 10,000€.
fewer than 5 players will find the winning numbers, winning 10,000€
'Fewer than 5 players will find the winning numbers, thus earning €10,000.'
=> some players will find the winning numbers

[28] As mentioned in Section 2.3.2, narrow scope ARCs might display a parasuppositional behavior. In addition, we explored the behavior of conjuncts with French *thus* ('ainsi'). As shown in (i), they yield clear projection patterns, but deviance under *not* and *none* is far less clear. We leave a more detailed investigation (and a comparison with *therefore* ('donc')) for future research.

(i) a. Est-ce qu' Anne voyage et prend ainsi des risques énormes?
is it that Anne travels and takes thus some risks énormes?
'Will Anne travel and thus take inconsiderate risks?'
=> if Anne travels, she takes inconsiderate risks
b. Peut-être qu' Anne voyage et prend ainsi des risques énormes.
maybe that Anne travels and takes thus some risks énormes.
'Maybe Anne travel and thus takes inconsiderate risks.'
=> if Anne travels, she takes inconsiderate risks

level, various aspects of our analysis will have to be refined, pertaining in particular to distributional restrictions beyond negative expressions, to discourse relations, and to the epistemic status of parasuppositions. Most importantly perhaps, one will have to understand why parasuppositions are triggered in the first place—what might be termed the Triggering Problem for parasuppositions.

Last, but not least, the diversity of presupposition-like phenomena in language (from presuppositions to parasuppositions through cosuppositions, and possibly including supplements and expressives) cries out for an explanation: what accounts for the existence of these phenomena as well as for their fine-grained differences?

Appendix I. Operators that Make Parasuppositions Eliminable

We ask below what kinds of operators make parasuppositions eliminable, and derive some initial results.

- *Propositional case*

We start by showing that propositional operators that make parasupposition triggers eliminable are like negation in turning truth to falsity. This means that, on the assumption that they are don't *always* yield falsity, thus must turn falsity to truth, and thus be identical to negation.

Result 1. A propositional operator Op satisfies (1) iff it satisfies (2)

(1) for all propositions p, q,
(Op [p & q] & [p => q]) <=> Op p

(2) if p = 1, Op p = 0

Proof.

If (1), (2):
Suppose p = 1. Pick q = 0. Then
[p => q] = 0
hence
(Op [p & q] & [p => q]) = 0
and thus
Op p = 0.

If (2), (1):
Suppose p = 0. Then p => q = 1 and (1) applied to p, q is trivially satisfied irrespective of the value of q.
Suppose p = 1. If q = 1, (1) is satisfied. If q = 0,
p => q = 0
and thus
(Op [p & q] & [p => q]) = 0.
Since Op p = 0, the equivalence in (1) is satisfied.

- *Quantificational case*

We show that generalized (conservative) quantifiers that behave like *No P* in making parasupposition triggers eliminable in their verbal argument must entail *No P*.

Note that the following result makes sense in the background of theories that predict universal projection under quantifiers, as is the case of Heim 1983 and Schlenker 2009, for instance.

Result 2. A conservative generalized quantifier of the form D P (for determiner D and restrictor P) satisfies (3) iff it satisfies (4)

(3) for all predicates Q, G, ([D P] [Q & G] & [Every P] (Q => G)) <=(>) [D P] Q
(4) for all predicates Q, [D P] Q => [No P] Q

Remark: The "=>" direction of the equivalence in (3) always follows from Conservativity and thus plays no role below:

Suppose [D P] [Q & G] & [Every P] (Q => G). By Conservativity, [D P] [P & Q & G]. Since [Every P] (Q => G), [P & Q & G] has the same extension as [P & Q], hence [D P] [P & Q], and by Conservativity again [D P] Q.

If (3), (4):

Suppose not (4). So [D P] Q, and for some d, d satisfies P and Q. Pick G so that d satisfies not G. Clearly, [Every P] (Q => G), and thus not (3).

If (4), (3):

By (4), from [D P] Q, we get [No P] Q, hence also [Every P] (Q => G)). By conservativity, from [D P] Q we get [D P] [P & Q], hence [D P] [P & Q & G], hence [D P] [Q & G].

Appendix II. Non-eliminability and Stalnaker's non-triviality conditions

As mentioned in the main text, Stalnaker's non-triviality conditions, copied in (111), look rather different from the (global) non-eliminability condition, copied in (112). We show below that (i) the Non-eliminability condition is not reducible to Stalnaker's non-triviality conditions, no matter which context they are evaluated with respect to, but that (ii) for our target expressions, one half of Stalnaker's conditions, namely (111)a, follows from the Non-eliminability condition.

(111) **Stalnaker's non-triviality conditions**
If E is an expression whose local context is c', then it should be the cas that:
a. c' $\not\models$ E
b. c' $\not\models$ not E

(112) **Non-eliminability condition (initial version)**
If … [P * G] … is asserted by a speaker whose belief state is S,

a. it should not (be common belief that) S $\models$ … [P * G] … <=> … P …
b. … unless G provides information that has some other role.

❑ *Non-eliminability cannot be replaced with Stalnaker's non-triviality*

One might think that the problem with parasuppositional expressions under *not* and *none* is that they are trivial relative to their local context, whether computed in the standard fashion or as a speaker's local context. But this idea is hopeless because it fails to draw the right distinctions: negation does not affect local context computation, but we must derive a contrast between the positive and the negative sentences in (113). The key is that the local contexts of the VP in these constructions are all the same, which means that local non-triviality conditions should affect them in the same way, contrary to fact.

(113) a. One / Every student [smokes, <thus> taking enormous risks].
b. One / Every student [smokes and thus takes enormous risks].
c. One / Every student [smokes (and takes enormous risks)].
a'. #Not one / No student [smoke, <thus> taking enormous risks].
b'. #Not one / No student [smoke and thus take enormous risks].
c'. #Not one / No student [smokes (and take enormous risks)].

One could object that *not one NP* should be analyzed non-compositionally as *no NP*, and note that presupposition projection facts have been argued to be different under existential and *none*-type quantifiers (Chemla 2009). But the same conclusion can be reached by comparing *every student VP* and *no student VP*, which have the same type of presupposition projection behavior (Chemla 2009).

❑ *Half of Stalnaker's non-triviality follows from Clausal non-eliminability*

While the truth-conditional part of Non-eliminability cannot be derived from Stalnaker's non-triviality conditions, half of the latter, namely condition (111)a, follows from Non-eliminability for our target constructions. In a nutshell, the argument is that if condition (111)a is violated, then a parasuppositional expression *G* is trivially true in its local context; but then it can be eliminated without changing the truth conditions obtained relative to the speaker's beliefs *S*. The detailed argument is laid out in (114).

(114) Suppose that the context set is C and the speaker's beliefs are S, and let be G be a parasuppositional expression appearing in an unembedded sentence
F_1 = [… [P * G] …], with a simpler alternative
F_2 = [… [P] …].
Let us call c' and s' the standard and the speaker's local contexts of [P, G] in F_1, and let us call c" and s" the standard and the speaker's local contexts of G in F_1.

We show the following:

If (i) c" $\models$ G, then (ii) it is common belief that [the speaker can assert F_1 <=> the speaker can assert F_2].

As a result, F_1 violates Non-eliminability because it provides the same information as F_2 but is more complex.

We note that (i) has the consequences in (iii) and in (iv).

(iii) $C \models F_1 \Leftrightarrow F_2$

(This follows by standard properties of local contexts: if relative to F_1 $c'' \models G$, $c' \models P \Rightarrow G$. Since c' is also the local of [P] in F_2 , (iv) follows.)

(iv) It is common belief that $S \models F_1 \Leftrightarrow F_2$

((iv) follows from (iii) from the definition of common belief in terms of iterations of speaker and addressee beliefs, see for instance Stalnaker 2002.)

Let us assume (i). Then it is common belief that:

• if the speaker can assert F_2, the speaker can assert F_1 (by (iv), and the fact that there are no pragmatic conditions on F_2 that are not already conditions on F_1)

• if the speaker can assert F_2, the speaker can assert F_1.

By (iv), if the speaker's beliefs support F_2, they support F_2. But we much check that the projection condition is satified, namely: $s'' \models G$. But since C is less specific than S, $c'' \models G$ imples $s'' \models G$, which satisfies the projection condition.

Appendix III. Complements on Embedded Conjunctions and Disjunctions

A. Embedded conjunctions analyzed without implicatures

In the main text, we outlined the generalizations in (115) below. Our analysis involved scalar implicatures. Here we show that even without scalar implicatures, parasupposition theory suffices to derive (115)b, but not (115)c. To derive the latter, the 'official' parasupposition must be strengthened for reasons familiar from presupposition theory when the Proviso Problem is taken into account.

(115) a. #[No N] [V * G]
b. [No N] [[V * G] and V']
c. [No N] [V and [V' * G]]

To explain why (115)a is acceptable, all we need to show is that the truth-conditional part of Non-eliminability in (49)a in the main text is satisfied because (116)a is not equivalent to (116)b when the parasupposition is taken into account. (116)a makes the contributions in (117).

(116) a. [No N] [[V * G] and V']
b. [No N] [V and V']

(117) Contributions of [No N] [[V G] and V']
a. Assertive condition: [No N] [[V and G] and V']
b. Projective condition: [Every N] [V => G]

Now it is clear that the parasuppositional expression in (116)a is stronger than its simpler alternative in (116)b. The reason is that the projective condition in (117)b yields the strong requirement that *[Every [N & V]] G*. Taking this requirement into account, the assertive condition in (117)a states that *[No N][[V and V']*, which doesn't suffice to make the projective condition vacuous (as there could be N's that satisfy V but not V').

More formally, suppose that an individual I satisfies N, V, not G, and not V', as outlined in (118). It is clear that in this situation (116)b is true, but not (116)a because i falsifies the Projective condition.

(118) A situation that makes (116)b appropriate and true but not (116)a
There is a single individual i satisfying N.
i also satisfies V, not G, and not V'.

As stated, this analysis makes the wrong predictions for (115)c. Once again, we must compare the contribution of (119)a to that of the simpler alternative in (119)b. With the analysis in (120), based on standard rules of presupposition projection, we end up predicting that the parasuppositional expression and its simpler alternative convey exactly the same information, as shown in (121).

(119) a. [No N] [V and [V' * G]]
b. [No N] [V and V']

(120) Contributions of [No N] [V and [V'* G]]
a. Assertive condition: [No N] [V and [V' and G]]
b. Projective condition: [Every N] [[V and V'] => G]

(121) a. (119)b => [(120)a and (120)b]
Suppose (119)b. Clearly, (120)a follows. (120)b also follows because every N fails to satisfies [V and V'] and thus vacuously satisfies the embedded material implication.
b. [(120)a and (120)b] => (119)b
Suppose, for contradiction, that (120)a and (120)b hold, but (119)b doesn't—and thus some N-individual

i satisfies V and V'. By (120)b, N also satisfies G, but this means that i refutes (120)a, since it satisfies N, V, V' and G.

But there might be a way out owing to an independent issue, called the Proviso Problem: in many cases, presuppositions that are predicted to be conditional in form are strengthened into unconditional ones (e.g. Geurts 1996, 1999, Lassiter 2012, Mandelkern 2016). Presupposition triggers that appear in the second position of a conjunction embedded under *none*-type quantifiers are no exception, as illustrated in (122).

(122) a. None of my students both has health problems and continues to smoke.
b. Predicted presupposition (e.g. Heim 1983): for each of my students x, if x has health problems, x has been smoking
c. Actual presupposition: for each of my students x, x has been smoking

In fact, the intuitive meaning of (123) yields a stronger inference than is predicted by standard rules of presupposition projection: we obtain the inference that *every student who smokes takes enormous risks* (or possibly something even stronger), not the weaker inference that *every student who cares about his health and smokes takes enormous risks*

(123)

Il n'	y	a	pas un	seul	étudiant	qui	à	la fois	se soucie	de
it NE	*there*	*have*	*not a*	*single*	*student*	*who*	*at*	*the time*	*SE worries*	*of*
sa	santé	et	fume ,	prenant	des	risques	énormes.			
his	*health*	*and*	*smokes,*	*taking*	*some*	*risks*	*enormous*.			

'There isn't a single student who both takes care of his health and smokes, thus taking enormous risks.'

When this strengthening is taken into account, the contributions of the parasuppositional quantified statement are as in (124), with the revised condition appearing boldfaced. This case then reduces to that in which the parasupposition is triggered in the first conjunct.

(124) Contributions of [No N] [V and [V' * G]]
a. Assertive condition: [No N] [V and [V' and G]]
b. Projective condition (with strengthening): **[Every N] [V' => G]**

We leave a full investigation of this alternative analysis for future research.

B. Embedded disjunctions

For embedded disjunctions, there is no implicature that could lead to the prediction that the Non-eliminability condition is satisfied (contrary to the case of embedded conjunctions discussed in Section 5.2). In addition, the analysis of Part A above applied to parasuppositional disjuncts also predicts that these should violate Non-eliminabilitty.

The reasoning is outlined in (125)-(126), and holds irrespective of whether Proviso-style strengthening is applied when the parasupposition trigger appears in the second disjunct.

(125) Contributions of [No N] [[V * G] or V']
a. Assertive condition: [No N] [[V and G] or V']
b. Projective condition: [Every N] [V => G]
c. Simpler alternative: [No N] [V or V']
d. Equivalence
(i) It is clear that [(a) & (b)] implies (c)
(ii) Suppose (c). It is immediate that (a). Furthermore, from (c) it follows that [No N] V, which means that (b) is vacuously satisfied. Therefore (c) implies [(a) & (b)].

(126) Contributions of [No N] [[V or [V' * G]]
a. Assertive condition: [No N] [[V or [V' and G]]
b. Projective condition (unstrengthened): [Every [N and not V] [V' => G]
b'. Projective condition (strengthened): [Every N] [V' => G]

c. Simple alternative: [No N] [V or V']
d. Equivalence (irrespective of whether one goes with (b) or with (b')]
(i) It is clear that [(a) & (b)/(b')] implies (c). Suppose that (a) and (b). Suppose, for contradiction, that some i satisfies N and [V or V']. i can't satisfy V because this would refute (a). Suppose i satisfies not V, and V'. By (b) i satisfies G. But this refutes (a), contra hypothesis. This shows that [(a) & (b)] implies (c). Since (b') is stronger than (b), it also follows that [(a) & (b')] implies (c).
(ii) It is also the case that (c) implies [(a) & (b)/(b')]. Assume (c). (a) immediately follows. Furthermore, from (c) it follows that [No N] V'. This means that (b') is vacuously satisfied. Thus (c) implies [(a) and (b'))]. Since (b') is stronger than (b), it also follows that (c) implies [(a) and (b)].

The facts with APPs arguably go in the expected direction, as illustrated in (127)-(128). Note that we use *ni* rather than *ou* below because the latter is arguably degraded in the immediate scope of *none*-type quantifiers. But the meaning of *[Aucun N][V ni V']* is indeed that of *[No N][V or V']*.

(127) a. #Aucun médecin ne fume, prenant des risques énormes.
no doctor NE smokes, taking some risks enormous.
b. #?Aucun médecin ne fume, prenant des risques énormes, ni ne boit.
no doctor NE smokes, taking some risks enormous, nor NE drinks.
c. #Aucun médecin ne boit ni ne fume, prenant des risques énormes.
no doctor NE drinks nor NE smokes, taking some risks enormous.
d. Certains médecins fument, prenant des risques énormes.
some doctors smoke, taking some risks enormous.
'Some doctors smoke, taking enormous risks.'

(128) Le jour de Kippour,
the day of Kippur,
'On Yom Kippur,
a. <#>aucun participant n' a bu, contrevenant aux ordres.[29]
no participant NE has drunk, contravening to-the orders.
b. <#> aucun participant n' a bu, contrevenant aux ordres, ni mangé.
no participant NE has drunk, contravening to-the orders, nor eaten.
c. <#> aucun participant n' a bu ni mangé, contrevenant aux ordres.
no participant NE has drunk nor eaten, contravening to-the orders.
d. certains participants ont bu, contrevenant aux ordres.
some participants have drunk, contravening to-the orders.
some participiants drank, violating orders.'

[29] Here there might be a marginal reading on which the APP is attached high, i.e. on which *the fact that no participant drank contravened to orders*—in other words, the order was that some people should drink. We leave this marginal reading for future research.

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